OneCare Clinic Management System

Software Requirement Specification

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CHAPTER 1: INTRODUCTION

The OneCare Clinic Integrated Pharmacy Management System is a digital solution designed to modernize clinic operations by integrating patient management, appointment scheduling, and pharmacy management into a single platform.

Currently, the clinic manages patient records manually while operating the pharmacy as a separate system. This fragmented approach causes delays, increases the risk of human error, and limits efficient inventory tracking.

The proposed system centralizes the management of patient records, appointments, prescriptions, and pharmacy operations, enabling improved efficiency, accuracy, and overall service quality.

1.1 Purpose

The purpose of this document is to provide a detailed description of the functional and non-functional requirements of the OneCare Clinic Integrated Pharmacy Management System.

This document serves as a reference for developers, stakeholders and the client to understand the system requirements before development begins.

Intended Audience

- Doctor
- Pharmacist
- System admin
- Admin
- Software developers
- Project supervisors

1.2 Scope

System Name: OneCare Clinic Integrated Pharmacy Management System

The following functionalities shall be provided by the system:

- Manage digital patient records
- Enable online appointment booking
- Maintain medicine inventory
- Generate prescriptions

- Provide automated stock updates
- Send appointment reminders
- Generate reports and analytics
- Provide chatbot support for patients

The following functionalities shall not be included by the system:

- Online payment processing
- Multi-clinic support
- Insurance claim processing
- Telemedicine consultation

1.3 Definitions, Acronyms, and Abbreviation

Table 1-1: Glossary of Terms

Term	**Definition**
Super admin	Super admin responsible for system management
Admin	Staff responsible for managing queues, appointments
Pharmacist	Staff responsible for dispensing medicines
Prescription	List of medicines issued by the doctor
Chatbot	Automated system responding to patient queries
Inventory	Stock of medicines available in pharmacy
SRS	Software Requirement Specification
API	Application Programming Interface

1.4 Document Overview

A structured specification of the software requirements for the OneCare Clinic Integrated Pharmacy Management System is provided in this document.

- An introduction to the system, including its purpose, scope, and key definitions, is presented in Chapter 1.
- The project scope and all relevant stakeholders are described in Chapter 2.

- The functional requirements of the system are defined in a structured and traceable format in Chapter 3.
- The non-functional requirements, aligned with ISO/IEC 25010 quality standards, are specified in Chapter 4.
- The business rules, system constraints, and underlying assumptions are outlined in Chapter 5.
- System models, including use case diagrams, use case descriptions, and activity diagrams, are presented in Chapters 6 to 8.
- The external interface requirements, including user, hardware, software, and communication interfaces, are defined in Chapter 9.
- The data requirements, including data entities, relationships, and integrity rules, are described in Chapter 10.
- Alternative solutions are discussed and their rejection is justified in Chapter 11.
- The feasibility of the system is evaluated from technical, economic, operational, and schedule perspectives in Chapter 12.
- The risk analysis and corresponding mitigation strategies are presented in Chapter 13.
- All references used in the preparation of this document are listed in Chapter 14.

CHAPTER 2: PROJECT SCOPE AND STAKEHOLDERS

2.1 Project Scope

The project aims to develop a web-based integrated system that improves clinical efficiency and pharmacy management.

Key goals:

- Reduce manual paperwork
- Improve patient experience
- Automate prescription processing
- Maintain accurate medicine inventory
- Provide management reports

Support for approximately 25 patients per day shall be provided by the system managed by one doctor and one pharmacy.

2.2 Stakeholders

Table 2-1: Stakeholders and Their Roles

Stakeholder	**Role**
Clinic Owner	Decision maker and project sponsor
Doctor	Creates prescriptions and manages patient records
Pharmacist	Dispenses medicine and manages stock
Super Admin	Manage system users, permissions, and reports
Admin	Manage queues, appointments
Patients	Book appointments and receive notifications
Development Team	Build and maintain the system

CHAPTER 3: FUNCTIONAL REQUIREMENTS

3.1 User & Access Management

Table 3-1 Functional Requirements Specification Table

ID	Description	Priority	Source	Status
FR-001	The system shall support the roles: Super Admin, Admin, Doctor, and Pharmacist, and access shall be controlled using a role-permission matrix.	High	System	Proposed
FR-002	User accounts shall be created, updated, deactivated, and assigned roles by Super Admin.	High	Super Admin	Proposed
FR-003	Role-based permissions shall be configurable and enforceable across all system modules.	High	Super Admin	Proposed
FR-004	Users shall be authenticated using a valid username and password.	High	System	Proposed
FR-005	Users shall be redirected to role-specific dashboards after successful authentication.	High	System	Proposed
FR-006	User accounts shall be locked for 15 minutes after five consecutive failed login attempts.	High	System	Proposed
FR-007	Password reset shall be performed with a time-limited email link.	High	System	Proposed
FR-008	User sessions shall be terminated after 30 minutes of inactivity.	High	System	Proposed

3.2 Clinic Configuration

Table 3-2: Super Admin Functional Requirements

ID	Description	Priority	Source	Status

FR-009	Doctor schedules including working days and time slots shall be configured by the Super Admin.	High	Super Admin	Proposed
FR-010	System data backup and restore operations shall be performed by Super Admin.	Medium	Super Admin	Proposed
FR-011	System-wide reports and audit logs shall be accessible to Super Admin.	High	Super Admin	Proposed

3.3 Patient Management

Table 3-3: Patient Management Functional Requirements

ID	Description	Priority	Source	Status
FR-012	Patient records shall be created by capturing personal and contact details.	High	Admin	Proposed
FR-013	Non-medical patient information shall be updated by Admin users.	High	Admin	Proposed
FR-014	Patient records shall be searchable using name, phone number, or clinic number.	High	Admin	Proposed

3.4 Appointment Management

Table 3-4: Appointment Management Functional Requirements

ID	Description	Priority	Source	Status
FR-015	Appointments shall be created by assigning patient, doctor, date, and time slot and preventing double booking.	High	Admin	Proposed
FR-016	Existing appointments shall be rescheduled.	High	Admin	Proposed
FR-017	Appointments shall be cancelled when required.	High	Admin	Proposed

FR-018	Appointment statuses shall be maintained as Scheduled, Completed, Cancelled, or No-show.	High	System	Proposed

3.5 Queue Management

Table 3-5: Queue Management Functional Requirements

ID	Description	Priority	Source	Status
FR-019	A daily patient queue shall be generated automatically based on appointments.	High	System	Proposed
FR-020	Queue status shall be updated when a consultation is marked as completed by the Doctor.	High	System	Proposed
FR-021	Patient arrival and consultation completion shall be recorded by Admin.	High	Admin	Proposed

3.6 Doctor Module

Table 3-6: Doctor Functional Requirements

ID	Description	Priority	Source	Status
FR-022	Scheduled appointments shall be viewable by the Doctor.	High	Doctor	Proposed
FR-023	Patient medical history and past consultations shall be accessible to the Doctor.	High	Doctor	Proposed
FR-024	Clinical notes shall be recorded during consultations.	High	Doctor	Proposed
FR-025	Digital prescriptions including medicine, dosage, duration, and instructions shall be created.	High	Doctor	Proposed
FR-026	Prescriptions shall be marked as “Pending Dispense” after creation.	High	System	Proposed

3.7 Pharmacy Module

Table 3-7: Pharmacy Functional Requirements

ID	Description	Priority	Source	Status
FR-027	Prescriptions with “Pending Dispense” status shall be viewable by Pharmacist.	High	Pharmacist	Proposed
FR-028	Medicines shall be dispensed after validating prescription details.	High	Pharmacist	Proposed
FR-029	Prescription status shall be updated as Fully Dispensed or Partially Dispensed.	High	System	Proposed
FR-030	Duplicate dispensing of completed prescriptions shall be prevented.	High	System	Proposed

3.8 Inventory Management

Table 3-8: Inventory Management Functional Requirements

ID	Description	Priority	Source	Status
FR-031	Medicine records shall be created, updated, and deactivated by Pharmacist.	High	Pharmacist	Proposed
FR-032	Stock levels shall be maintained and updated after dispensing.	High	Pharmacist	Proposed
FR-033	Expiry dates shall be tracked and expired medicines shall be marked unavailable.	High	Pharmacist	Proposed
FR-034	Low stock alerts shall be generated when thresholds are reached.	Medium	Pharmacist	Proposed

3.9 Billing

Table 3-9: Billing Functional Requirements

ID	Description	Priority	Source	Status
FR-035	Invoices shall be generated including consultation and medicine charges.	High	Admin	Proposed

FR-036	Only cash payments shall be accepted as per BR-011.	High	Admin	Proposed
FR-037	Invoice status shall be maintained as Paid or Pending.	High	Admin	Proposed
FR-038	Historical invoices shall be retrievable.	Medium	Admin	Proposed

3.10 Reporting & Audit

Table 3-10: Reporting & Audit Functional Requirements

ID	Description	Priority	Source	Status
FR-039	Clinic reports including patient counts and consultations shall be generated.	Medium	System	Proposed
FR-040	Pharmacy reports including stock usage and expiry shall be generated.	Medium	Pharmacist	Proposed
FR-041	User activities including login, data changes, and dispensing actions shall be logged.	High	System	Proposed
FR-042	Audit logs shall be viewable with filtering capabilities.	High	Super Admin	Proposed

3.11 Patient Self-Service & Chatbot

Table 3-11: Patient Self-Service Functional Requirements

ID	Description	Priority	Source	Status
FR-043	Appointments shall be booked by patients without logging in using basic personal details.	High	Patient	Proposed
FR-044	A chatbot interface shall be provided for querying clinic details such as schedules and services.	Medium	Patient	Proposed

FR-045	Chatbot responses to medicine-related queries shall be limited to predefined approved information.	Medium	System	Proposed
FR-046	Health-related responses shall be generated only from doctor-approved PDF knowledge sources.	Medium	Doctor	Proposed
FR-047	PDF documents used for chatbot knowledge shall be uploaded and managed by authorized users.	Medium	Doctor	Proposed

3.12 External Dispensing Module

Table 3-12: Functional Requirements for External Dispensing and Delivery Management

ID	Requirement	Priority	Source	Status
FR-048	The system shall allow Pharmacist to mark a prescription for external dispensing to an off-site patient.	High	Pharmacist	Proposed
FR-049	The system shall require verification of patient identity using phone number, clinic ID, or QR code before dispensing medicines externally.	High	Pharmacist	Proposed
FR-050	External dispensing shall update the inventory immediately and mark the prescription status as “Dispensed (External)”.	High	System	Proposed
FR-051	A delivery record shall be generated including patient ID, staff ID, date/time, and payment method.	High	Pharmacist	Proposed
FR-052	The system shall maintain audit logs for all external dispensing activities.	High	Compliance	Proposed

CHAPTER 4: NON-FUNCTIONAL REQUIREMENTS

4.1 Performance

Table 4-1: Performance Requirements and Acceptance Criteria

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-P-001	Dashboard Performance	Dashboard pages shall be loaded within ≤3 seconds for 95% of requests under ≤50 concurrent users.	High	Admin	Proposed
NFR-P-002	API Performance	95% of API responses shall be returned within ≤2 seconds.	High	Dev Team	Proposed
NFR-P-003	Booking Performance	Appointment booking shall be completed within ≤2 seconds.	High	Patient	Proposed
NFR-P-004	Queue Update Latency	Queue updates shall be reflected within ≤2 seconds after consultation completion is recorded.	High	Staff	Proposed
NFR-P-005	Notification Time	SMS notifications shall be sent within ≤10 seconds of triggering events.	High	Patient	Proposed
NFR-P-006	Report Generation	Reports up to 1 year of data shall be generated within ≤10 seconds.	Medium	Admin	Proposed

4.2 Reliability

Table 4-2: Reliability Non-Functional Requirements

ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-R-001	Availability	System uptime shall be ≥99% per month excluding maintenance.	High	Admin	Proposed
NFR-R-002	Offline Capability	Queue updates, appointment logging, and billing shall be allowed to be performed for up to 5 minutes during offline operations; data shall be synchronized automatically after connectivity is restored.	High	Staff	Proposed
NFR-R-003	Data Sync	Data created during offline operation shall be automatically synchronized with the central database within 5 minutes after connectivity is restored.	High	Dev Team	Proposed
NFR-R-004	Backup	Automated backups shall be performed every 24 hours.	High	Admin	Proposed
NFR-R-005	Recovery	System recovery shall be completed within ≤4 hours after failure.	High	Management	Proposed

4.3 Usability

Table 4-3: Usability Requirements

NFR ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-U-001	Multilingual interface	English and Tamil shall be supported by the system interface for all patient-facing pages.	High	Staff	Proposed
NFR-U-002	Ease of learning	Basic tasks (patient registration, queue updates, and invoice generation) shall be able to be performed by a new staff member within 30 minutes of initial use.	High	Staff	Proposed
NFR-U-003	Task efficiency	Common operational tasks (confirm booking, update stock, and generate report) shall be required to be completed within no more than three navigation steps.	High	Admin	Proposed
NFR-U-004	Mobile responsiveness	Correct functioning shall be ensured by the system interface on desktop, tablet, and mobile devices with screen widths ≥360px.	High	Staff / Patients	Proposed
NFR-U-005	Form validation	Invalid form inputs shall be detected, and inline validation messages shall be displayed before submission.	Medium	Staff	Proposed

4.4 Security

Table 4-4: Security Non-Functional Requirements

NFR ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-S-001	Password policy	Password complexity rules shall be met, and passwords shall be securely hashed.	High	Admin	Proposed
NFR-S-002	Session management	Automatic expiration of user sessions shall occur after 30 minutes of inactivity.	High	Dev Team	Proposed
NFR-S-003	Role-based access control	Role-Based Access Control (RBAC) shall be enforced by the system to restrict access to authorized modules only.	High	Admin	Proposed
NFR-S-004	Secure communication	All client-server communications shall be encrypted using HTTPS with TLS 1.2 or higher.	High	Dev Team	Proposed
NFR-S-005	Data protection	Patient and prescription data shall be accessible only to authorized users according to defined access permissions.	High	Compliance	Proposed
NFR-S-006	Audit logging	Logs for critical actions, including user login, patient record updates, prescription creation, and	High	Admin	Proposed

		stock changes, shall be recorded by the system.			
NFR-S-007	Input validation	All user inputs shall be validated to prevent SQL injections, cross-site scripting (XSS), and other injection attacks.	High	Dev Team	Proposed
NFR-S-008	CSRF Protection	Cross-Site Request Forgery (CSRF) protection shall be implemented by the system for all state-changing HTTP requests (POST, PUT, DELETE).	High	Dev Team	Proposed
NFR-S-009	Rate Limiting	Authentication requests shall be limited by the system to a maximum of 10 attempts per minute per IP address to prevent brute-force attacks.	High	Dev Team	Proposed
NFR-S-010	Data Encryption at Rest	Sensitive data, including patient records and authentication credentials, shall be encrypted at rest using industry-standard encryption (e.g., AES-256).	High	Compliance	Proposed

4.5 Scalability

Table 4-5: Scalability Non-Functional Requirements

NFR ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-SC-001	Concurrent user capacity	Support for at least 50 concurrent users shall be ensured by the system without exceeding the response times defined in NFR-P-001 and NFR-P-002.	High	Dev Team	Proposed
NFR-SC-002	Database growth	Support for up to 20,000 patient records and 5 years of transaction history shall be provided by the system database without performance degradation.	High	Dev Team	Proposed

4.6 Maintainability

Table 4-6: Maintainability Non-Functional Requirements

NFR ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-M-001	Modular architecture	A modular architecture separating frontend, backend, and database layers shall be followed by the system.	High	Dev Team	Proposed
NFR-M-002	API documentation	All backend APIs shall be documented using Open API / Swagger standards.	High	Dev Team	Proposed

NFR-M-003	Code quality	At least 70%-unit test coverage shall be achieved for the backend source code.	Medium	Dev Team	Proposed
NFR-M-004	Configuration management	Environment-specific configurations shall be stored outside the source code using environment variables or secure configuration files.	High	Dev Team	Proposed
NFR-M-005	Logging	Application logs shall be stored in a centralized logging system for a minimum of 90 days.	Medium	Dev Team	Proposed

4.7 Portability

Table 4-7: Portability Non-Functional Requirements

NFR ID	Requirement	Measurable Acceptance Criterion	Priority	Source	Status
NFR-PO-001	Browser compatibility	Correct functioning shall be ensured by the web application on Chrome, Firefox, Edge, and Safari browsers.	High	Dev Team	Proposed
NFR-PO-002	Container deployment	The application shall be deployable using Docker containers.	High	Dev Team	Proposed
NFR-PO-003	Operating system independence	Operation shall be supported by the system on any operating system that supports Docker containers.	High	Dev Team	Proposed

NFR-PO-004	Database portability	The database schema shall be deployable on MySQL version 8.0 or higher.	High	Dev Team	Proposed
NFR-PO-005	Language extensibility	Additional languages should be provided using separate translation files without modifying the application source code.	Low	Dev Team	Proposed

CHAPTER 5: BUSINESS RULES, CONSTRAINTS AND ASSUMPTIONS

5.1 Business Rules

Table 5-1: Business Rules Description

ID	Rule Description
BR-001	Defined complexity rules shall be complied with by user passwords, and secure hashing mechanisms shall be used for their storage.
BR-002	User accounts shall be temporarily locked after a defined number of consecutive failed login attempts.
BR-003	Each patient shall be assigned a unique patient identifier within the system.
BR-004	Doctors shall be permitted to create prescriptions but shall not be permitted to modify pharmacy inventory records.
BR-005	Pharmacist shall be permitted to dispense medicines but shall not be permitted to modify patient medical records.
BR-006	Administrative users shall be restricted to accessing and modifying non-clinical patient information only.
BR-007	Super Admin users shall be responsible for managing user accounts, permissions, and system-level reports.
BR-008	Password reset operations for system users shall be performed only by the Super Admin, and users shall be required to change the password upon first login after reset.
BR-009	Chatbot responses shall be restricted to general clinic information and predefined health knowledge approved by a qualified doctor.
BR-010	Chatbot shall not provide diagnosis, prescription, or medical advice beyond approved informational content.
BR-011	All patient payments shall be made in cash; no electronic or card payments shall be supported by the system

5.2 Constraints

5.2.1 Technology Constraints

Table 5-2: Technology Constraints

ID	Constraint
CON-001	The system shall be implemented as a web-based application accessible via modern web browsers (Chrome, Edge, Firefox, Safari).
CON-002	The system shall be compatible with Windows-based systems used within the clinic environment.

5.2.2 Legal Constraints

Table 5-3: Legal Constraints

ID	Constraint
CON-003	Patient medical data shall be handled in compliance with applicable health data privacy regulations.
CON-004	Access to patient medical records shall be restricted to authorized medical personnel only.
CON-005	Patient data shall be securely stored and retained in accordance with defined medical record retention policies.

5.2.3 Operational Constraints

Table 5-4: Operational Constraints

ID	Constraint
CON-006	Operations for a single-doctor clinic environment shall be supported by the system.
CON-007	Operational continuity shall be maintained by the system under moderate internet connectivity conditions.

5.3 Assumptions

Table 5-5: System Assumptions

ID	Assumption Description
ASM-001	Staff are assumed to possess basic computer literacy to operate the system effectively.
ASM-002	Existing patient records and medicine inventory are assumed to be manually entered into the system during the initial setup phase.
ASM-003	Reliable internet connection is assumed to be available for accessing the cloud-hosted system.
ASM-004	Regular system backups are assumed to be performed to safeguard patient and pharmacy data.
ASM-005	Adequate computer devices are assumed to be provided for all staff required to operate the system.

CHAPTER 6: USE CASE DIAGRAMS

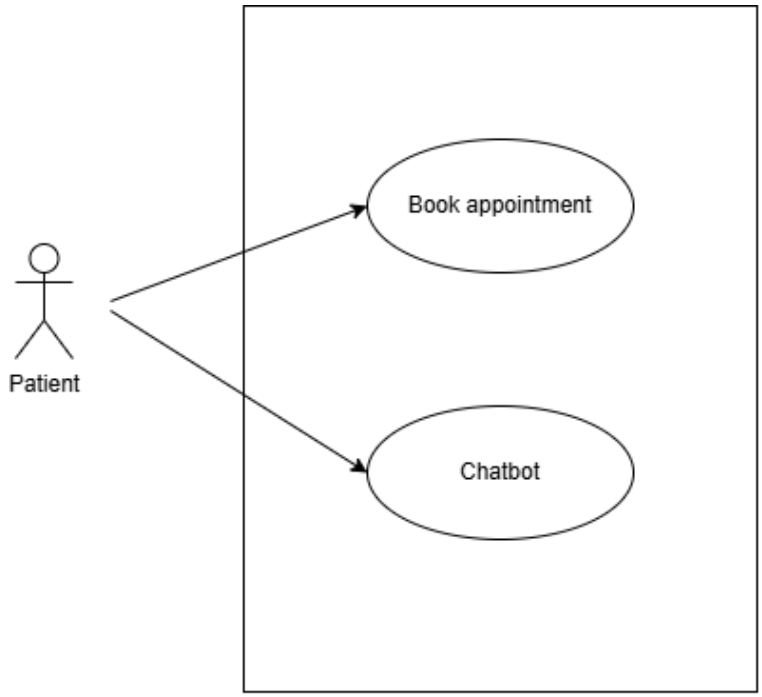


Figure 6-1: Use Case Diagram for Patient

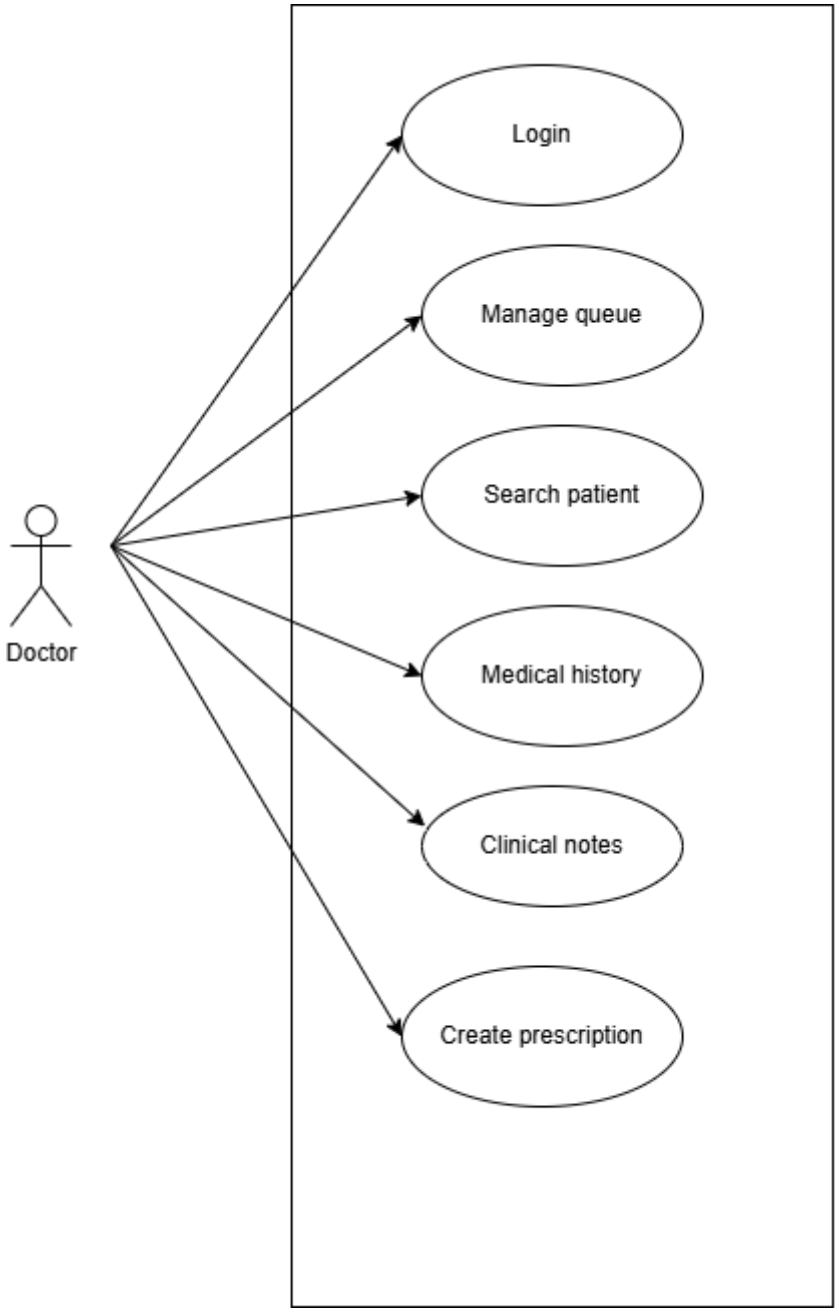


Figure 6-2: Use Case Diagram for Doctor

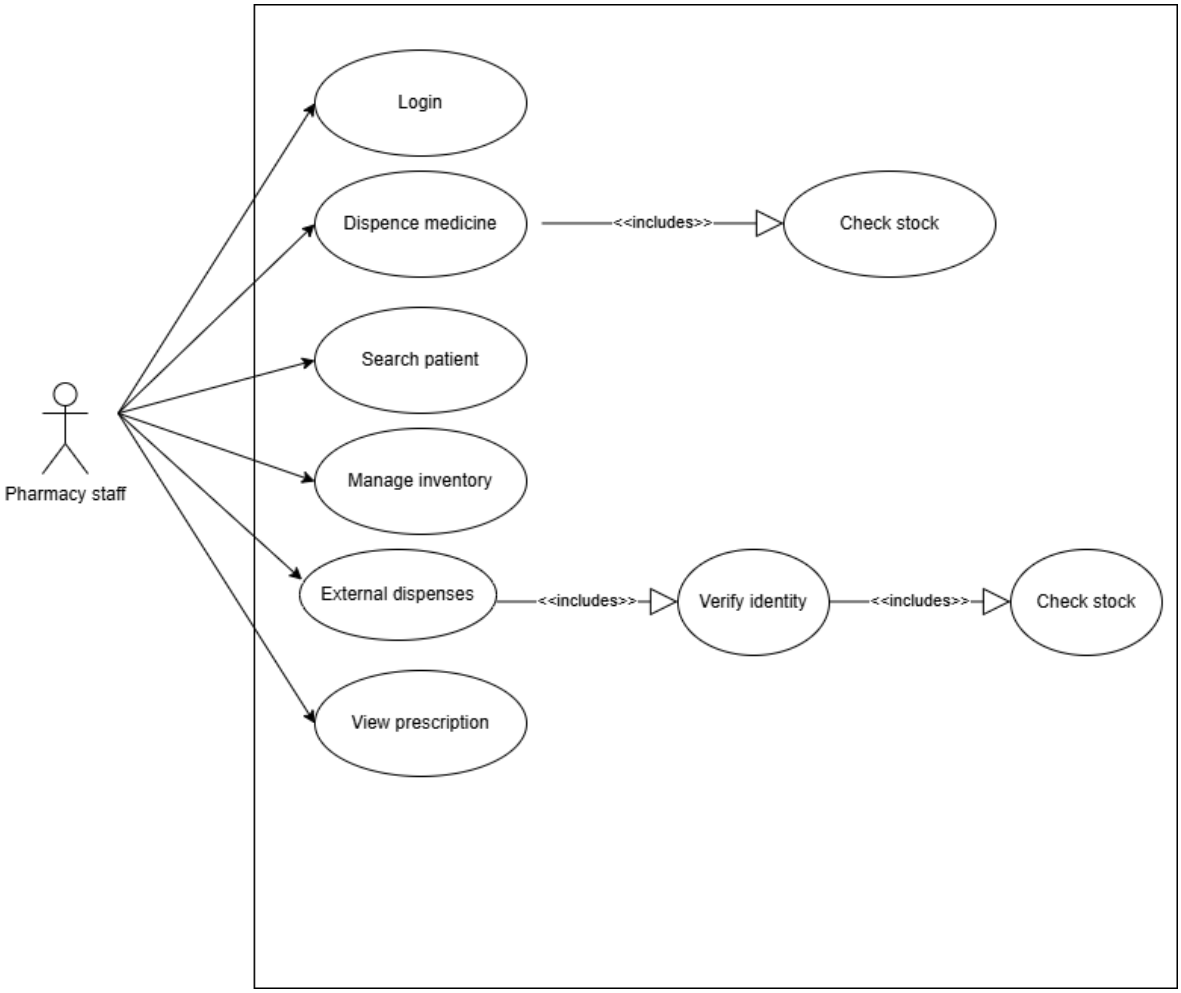


Figure 6-3: Use Case Diagram for Pharmacy Staff

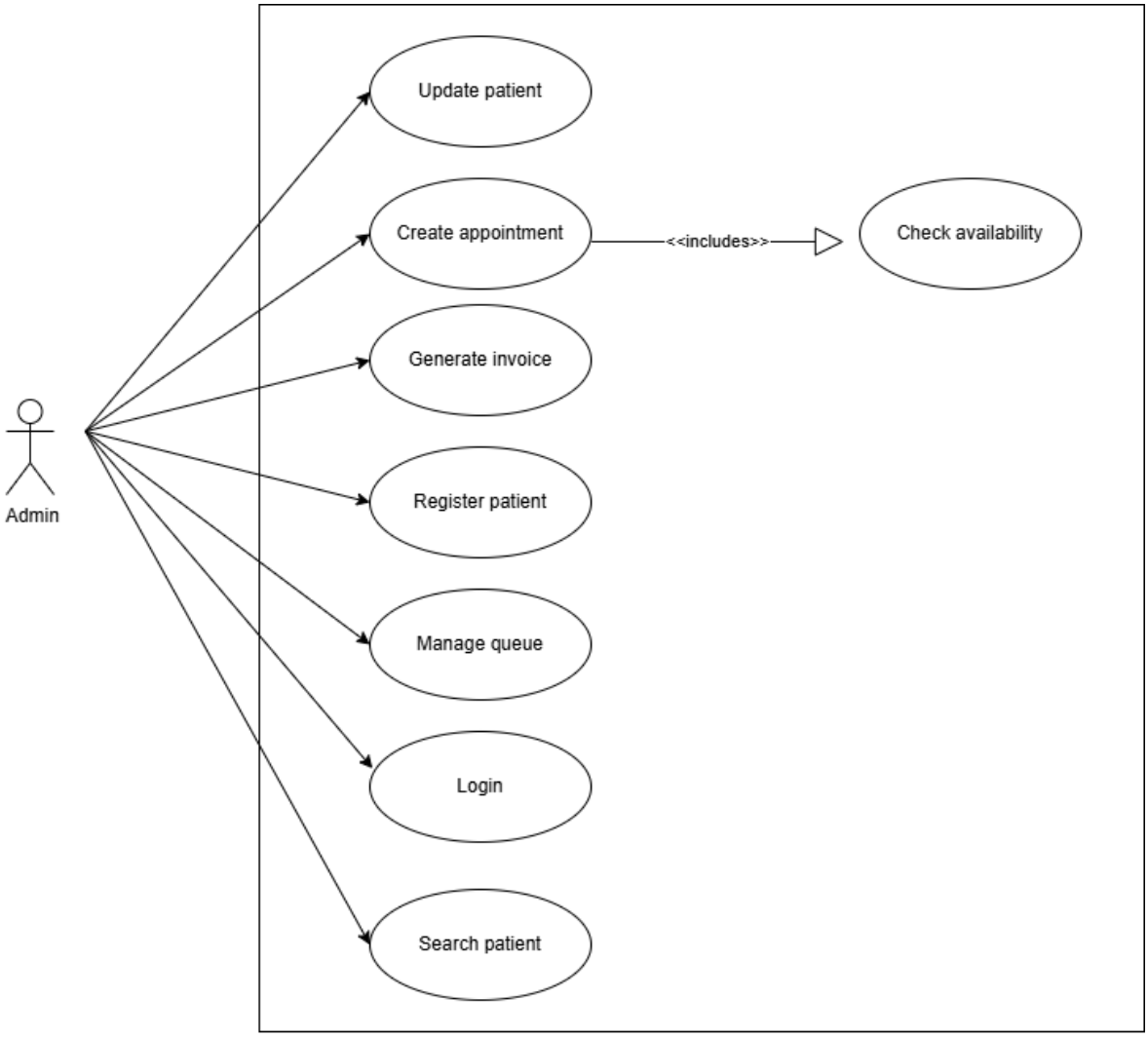


Figure 6-4: Use Case Diagram for Admin

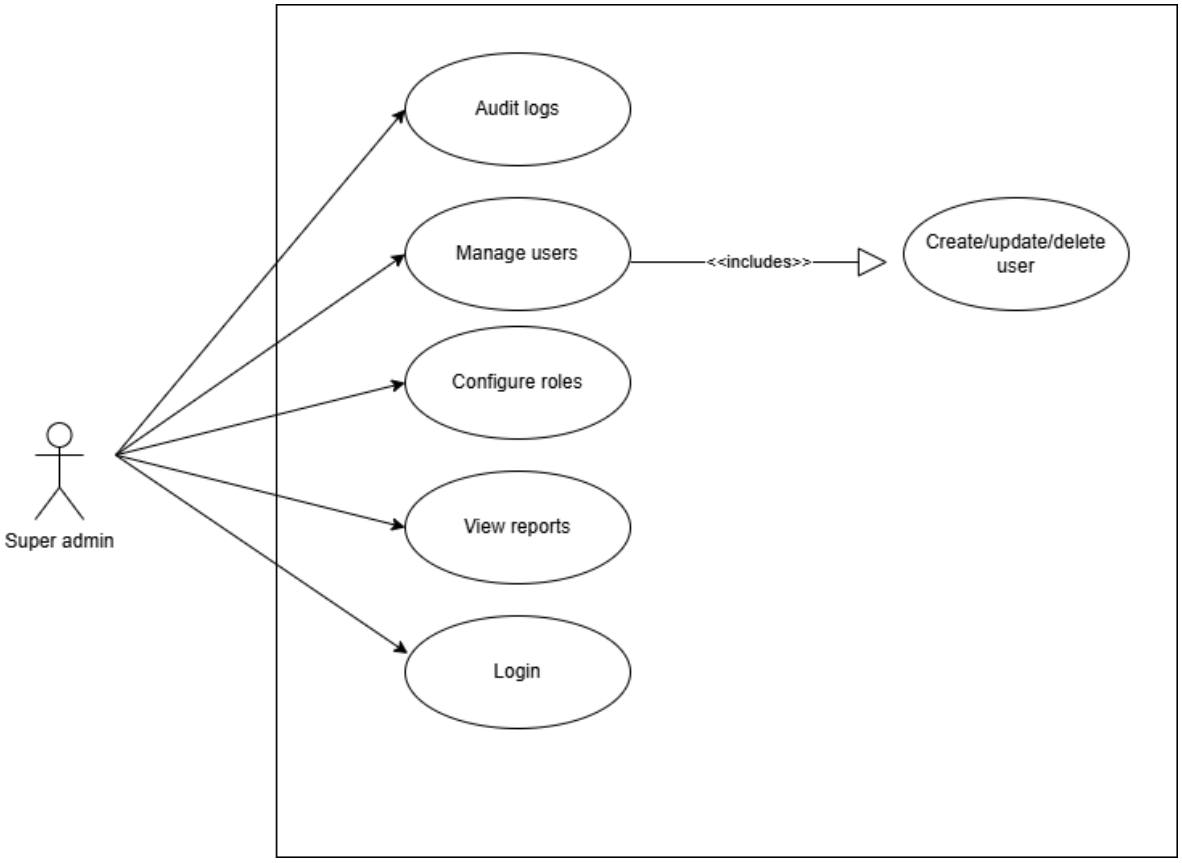


Figure 6-5: Use Case Diagram for Super Admin

CHAPTER 7: USE CASE DESCRIPTIONS

The following case descriptions define the interaction sequences between system actors and the OneCare Clinic Management System. Each use case is derived from the validated scenario specifications agreed upon during requirements elicitation.

Table 7-1: Use case of User Login

Use Case ID	1
Use Case Name	User Login
Description	Handles secure authentication of users (Doctor, Admin, Pharmacy Staff, Super Admin). It validates credentials, manages sessions, logs activities, and ensures account security
Actor	Doctor, Pharmacy Staff, Admin, Super Admin
Pre-Conditions	A user account exists in the system. The user possesses valid credentials.
Post Conditions	The authenticated user accesses the role-specific dashboard. A login event is recorded in the audit log with a timestamp and client IP address. A session with a 30-minute inactivity timeout is active.
Main Flow	1. The user navigates to the login page. 2. The system prompts the user to enter their username/email and password. 3. The system validates the credentials. 4. Upon successful validation, the user is logged in and redirected to their dashboard.
Alternative Flows	Invalid credentials: The system displays: 'Invalid username or password.' After five consecutive failed attempts, the account is locked for 15 minutes. The system logs the lockout event in the audit log. The system displays a lockout message with unlock instructions.

Table 7-2: Use case of Manage user accounts

Use Case ID	2
Use Case Name	Manage user accounts
Description	Allows the Super Admin to create, update, deactivate, and reset user accounts. Ensure proper user lifecycle management and maintain audit logs.
Actor	Super Admin
Pre-Conditions	The Super Admin is authenticated and on the User Management module
Post Conditions	The user account is created, updated, or deactivated as requested. The relevant audit log entry is recorded.
Main Flow	1. The Super Admin navigates to the User Management section. 2. The system displays a list of existing user accounts. 3. The Super Admin selects Create New User. 4. The system displays the user creation form. 5. The Super Admin enters the user's name, email, role, and initial password. 6. The Super Admin submits the form. 7. The system validates the input and creates the account. 8. The system sends a welcome email prompting the user to change the password on first login. 9. The system logs the account creation event in the audit log.
Alternative Flows	Update User: The Super Admin selects an existing user and clicks Edit. The Super Admin modifies allowable fields (name, role, status). The Super Admin saves the changes. The system updates the record and logs the change. Deactivate User: The Super Admin selects the user and clicks Deactivate. The system prompts confirmation. Upon confirmation, the account is deactivated and the user is prevented from logging in. The system logs the deactivation event.

	Reset Password: The Super Admin selects the user and clicks Reset Password. The system generates a temporary password or sends a reset link. The user is required to change the password on next login. The system logs the password reset event

Table 7-3: Use case of Configure Role-Based Permissions

Use Case ID	3
Use Case Name	Configure Role-Based Permissions
Description	Enables the Super Admin to assign and control access permissions for different roles, ensuring secure and role-based system access.
Actor	Super Admin
Pre-Conditions	The Super Admin is authenticated. Roles exist in the system (Super Admin, Admin, Doctor, Pharmacy Staff).
Post Conditions	Updated permissions are enforced across all system modules for the selected role. The permission change is recorded in the audit log.
Main Flow	1. The Super Admin navigates to the Role & Permissions configuration section. 2. The system displays the role-permission matrix. 3. The Super Admin selects a role to configure. 4. The system displays the current permissions assigned to the role. 5. The Super Admin enables or disables specific module permissions. 6. The Super Admin saves the configuration. 7. The system applies the updated permissions and logs the change in the audit log
Alternative Flows	The Super Admin reviews permissions without modifying. The Super Admin cancels, and no changes have persisted

Table 7-4: Use case of Configure Doctor Schedule

Use Case ID	4
Use Case Name	Configure Doctor Schedule
Description	Allow the Super Admin to define doctors' working days and time slots, making them available for appointment booking.
Actor	Super Admin
Pre-Conditions	The Super Admin is authenticated. Doctor user accounts exist in the system.
Post Conditions	The doctor's schedule is saved and available for appointment creation. The configuration change is recorded in the audit log
Main Flow	1. The Super Admin navigates to the Clinic Configuration section. 2. The Super Admin selects the doctor whose schedule is to be configured. 3. The system displays the current schedule of the selected doctor. 4. The Super Admin sets working days and time slots. 5. The Super Admin saves the schedule. 6. The system validates the configuration and persists in the schedule. 7. The system makes the doctor's schedule available for appointment booking
Alternative Flows	The system detects a conflict in the defined time slots. The system displays a validation error indicating conflict. The Super Admin corrects the time slot and resubmits

Table 7-5: Use case of Backup and Restore System Data

Use Case ID	5
Use Case Name	Backup and Restore System Data
Description	Ensures data safety by allowing system backups and restoration in case of failures or data loss.
Actor	Super Admin
Pre-Conditions	The Super Admin is authenticated.

	The system is operational
Post Conditions	A backup file is created and stored, or data is restored from a selected backup. The backup or restore event is logged in the audit log.
Main Flow	1. The Super Admin navigates to the System Administration section. 2. The Super Admin selects Backup Data. 3. The system initiates a backup of all system data. 4. The system confirms successful backup completion and records the timestamp. 5. The Super Admin can download or store the backup file securely
Alternative Flows	The Super Admin selects Restore Data. The Super Admin uploads or selects a backup file. The system prompts for confirmation and warning of data overwritten. Upon confirmation, the system restores data from the backup. The system confirms successful restoration

Table 7-6: Use case of View System-Wide Reports and Audit Logs

Use Case ID	6
Use Case Name	View System-Wide Reports and Audit Logs
Description	Provides access to system reports and logs for monitoring, auditing, and decision-making.
Actor	Super Admin
Pre-Conditions	The Super Admin is authenticated. System activity data and audit logs exist.
Post Conditions	The requested report or audit log is displayed. Exported files are available for download.
Main Flow	1. The Super Admin navigates to the Reports & Audit Logs section. 2. The system displays report categories and audit log filters.

	3. The Super Admin selects the report type or applies filters (date range, user, action type). 4. The system generates and displays the requested report or filtered audit log. 5. The Super Admin can export the report or log as a file
Alternative Flows	The system returns no results for the applied filters. The system displays: 'No records found for the selected criteria.' The Super Admin adjusts the filters and retries

Table 7-7: Use case of Register new patient

Use Case ID	7
Use Case Name	Registering New Patient
Description	Enables Admin to create new patient records with unique clinic IDs.
Actor	Admin
Pre-Conditions	The admin is authenticated. The patient is not already registered in the system.
Post Conditions	A new patient record is created with a unique clinic number. The patient is searchable in the system
Main Flow	1. The admin navigates to the Patient Management module. 2. The admin selects Register New Patient. 3. The system displays the patient registration form. 4. The admin enters the patient's personal and contact details. 5. The admin submits the form. 6. The system validates the input fields. 7. The system assigns a unique clinic number (patient ID) to the patient. 8. The system creates the patient record and saves it. 9. The system displays confirmation with the assigned clinic number

Alternative Flows	The system detects a potential duplicate based on name and phone number. The system warns the admin of the potential duplicate. The admin reviews the existing record and either proceeds with creation or cancels.

Table 7-8: Use case of Update Patient Information

Use Case ID	8
Use Case Name	Update Patient Information
Description	Allows Admin to update patient details such as contact information and address.
Actor	Admin
Pre-Conditions	The admin is authenticated. The patient record exists in the system
Post Conditions	The patient's non-medical information is updated. The update is recorded in the audit log.
Main Flow	1. The admin searches for the patient by name, phone number, or clinic number. 2. The system displays matching patient record. 3. The admin selects the patient and clicks Edit. 4. The system displays editable non-medical fields (contact details, address, etc.). 5. The admin modifies the required fields. 6. The admin saves the changes. 7. The system validates and persists the updated record. 8. The system logs the update in the audit log
Alternative Flows	No matching patient record is found. The system displays: 'No patient found matching the search criteria.' The admin refines the search or registers the patient

Table 7-9: Use case of Search for Patient Records

Use Case ID	9

Use Case Name	Search for Patient Records
Description	Allows users (Admin, Doctor, Pharmacy Staff) to quickly locate patient records using search criteria.
Actor	Admin, Doctor, Pharmacy Staff
Pre-Conditions	The user is authenticated. Patient records exist in the system.
Post Conditions	The patient record is located and displayed to the user
Main Flow	1. The user navigates to the Patient Search section. 2. The user enters a search term (name, phone number, or clinic number). 3. The user submits the search. 4. The system queries the database and returns matching results. 5. The user selects the desired patient record from the results. 6. The system displays the patient's details appropriate to the user's role
Alternative Flows	The system returns no results. The system displays: 'No matching patient records found.' The user revises the search term or registers the patient

Table 7-10: Use case of Create Appointment

Use Case ID	10
Use Case Name	Create Appointment
Description	Allows Admin to schedule appointments by selecting patient, doctor, and available time slots.
Actor	Admin
Pre-Conditions	The admin is authenticated. The patient record exists. The doctor's schedule is configured
Post Conditions	The appointment is created with status 'Scheduled'. The appointment appears in the daily queue.

Main Flow	1. The admin navigates to the Appointment Management module. 2. The admin selects Create New Appointment. 3. The system displays the appointment creation form. 4. The admin searches for and selects the patient. 5. The admin selects the doctor, date, and available time slot. 6. The admin submits the form. 7. The system validates availability and absence of conflicts. 8. The system creates the appointment with status 'Scheduled'. 9. The system adds the appointment to the doctor's schedule. 10. The system sends an appointment confirmation notification to the patient
Alternative Flows	The selected time slot is already booked. The system displays available alternative slots. The admin selects an available slot and resubmits.

Table 7-11: Use case of Reschedule Appointment

Use Case ID	11
Use Case Name	Reschedule Appointment
Description	Enables Admin to modify existing appointment dates and times.
Actor	Admin
Pre-Conditions	The admin is authenticated. An existing appointment with status 'Scheduled' exists
Post Conditions	The appointment is updated with the new date and time. The patient receives a reschedule notification
Main Flow	1. The admin searches for and selects the existing appointment. 2. The admin clicks Reschedule.

	3. The system displays available time slots for the same or different date. 4. The admin selects a new date and time slot. 5. The admin confirms the reschedule. 6. The system updates the appointment with the new date and time. 7. The system notifies the patient of the updated appointment details
Alternative Flows	No available slots exist for the requested date. The system prompts the admin to select a different date

Table 7-12: Use case of Cancel Appointment

Use Case ID	12
Use Case Name	Cancel Appointment
Description	Allow Admin to cancel scheduled appointments and free up time slots.
Actor	Admin
Pre-Conditions	The admin is authenticated. An existing appointment with status 'Scheduled' exists.
Post Conditions	The appointment status is set to 'Cancelled'. The time slot is available for new bookings
Main Flow	1. The admin searches for and selects the appointment. 2. The admin clicks Cancel Appointment. 3. The system prompts cancellation confirmation. 4. The admin confirms the cancellation. 5. The system updates the appointment status to 'Cancelled'. 6. The system releases the time slot for other bookings. 7. The system notifies the patient of the cancellation
Alternative Flows	The appointment status is 'Completed' or 'No-show'. The system prevents cancellation and displays an appropriate message

Table 7-13: Use case of View and Manage Daily Queue

Use Case ID	13
Use Case Name	View and Manage Daily Queue
Description	Manages daily patient flow by organizing appointments into a queue and tracking statuses (Arrived, Completed, No-show)..
Actor	Admin, Doctor
Pre-Conditions	The user is authenticated. Appointments are scheduled for the current day.
Post Conditions	Queue statuses are updated in real time. Completed consultations are reflected in the patient's history
Main Flow	1. The system automatically generates the daily patient queue based on scheduled appointments at the start of each day. 2. The Admin or Doctor navigates to the Queue Management module. 3. The system displays the ordered list of patients with their queue number and appointment details. 4. The admin marks a patient as 'Arrived' when the patient checks in. 5. The Doctor selects the next patient in the queue to begin consultation. 6. Upon completing the consultation, the Doctor marks the consultation as 'Completed'. 7. The system updates the queue status and advances to the next patient. 8. The system records the consultation completion timestamp
Alternative Flows	The patient does not arrive within the expected timeframe. The admin marks the appointment status as 'No-show'. The system moves to the next patient in the queue

Table 7-14: Use case of View Scheduled Appointments

Use Case ID	14
Use Case Name	View Scheduled Appointments

Description	Allow doctor to view and filter their appointments.
Actor	Doctor
Pre-Conditions	The Doctor is authenticated. Appointments are scheduled for the current or future dates
Post Conditions	The Doctor views the scheduled appointments and can proceed to consultation
Main Flow	1. The Doctor navigates to the Appointments section of the dashboard. 2. The system displays all appointments scheduled for the current day by default. 3. The Doctor can filter appointments by date range. 4. The system displays patient name, appointment time, and queue status. 5. The Doctor selects an appointment to view detailed patient information
Alternative Flows	No appointments exist for the selected date. The system displays: 'No appointments scheduled for the selected date.'

Table 7-15: Use Case Access Patient Medical History

Use Case ID	15
Use Case Name	Access Patient Medical History
Description	Enables doctors to review past consultations, notes, and prescriptions.
Actor	Doctor
Pre-Conditions	The Doctor is authenticated. The patient record exists with prior consultation history
Post Conditions	The Doctor has reviewed the patient's medical history
Main Flow	1. The Doctor selects a patient from the appointment queue or via search. 2. The system displays the patient's profile. 3. The Doctor selects the Medical History tab.

	4. The system retrieves and displays all past consultations, clinical notes, and prescriptions. 5. The Doctor reviews the history to inform the current consultation
Alternative Flows	The patient has no prior consultations. The system displays: 'No medical history on record for this patient’

Table 7-16: Use case of Record Clinical Notes

Use Case ID	16
Use Case Name	Record Clinical Notes
Description	Allow doctors to document diagnosis, symptoms, and treatment plans.
Actor	Doctor
Pre-Conditions	The Doctor is authenticated. A consultation session is in progress for the selected patient.
Post Conditions	Clinical notes are saved and associated with the patient's consultation record. The notes are accessible in future medical history reviews.
Main Flow	1. The Doctor opens the active consultation for the patient. 2. The system displays the clinical notes entry form. 3. The Doctor enters symptoms, examination findings, diagnosis, and treatment plan. 4. The Doctor saves the clinical notes. 5. The system stores the notes linked to the current consultation record. 6. The system timestamps the record with the date and doctor ID
Alternative Flows	The Doctor decides not to save the notes. The Doctor clicks Discard. The system prompts confirmation. Upon confirmation, the unsaved notes are discarded

Table 7-17: Use case of Create Digital Prescription

Use Case ID	17
Use Case Name	Create Digital Prescription
Description	Enables doctors to generate prescriptions linked to consultations and send them to the pharmacy.
Actor	Doctor
Pre-Conditions	The Doctor is authenticated. A consultation session is in progress. Medicine records exist in the inventory.
Post Conditions	The prescription is saved with status 'Pending Dispense'. The Pharmacy Staff can view the prescription for dispensing.
Main Flow	1. During an active consultation, the Doctor selects Create Prescription. 2. The system displays the prescription form. 3. The Doctor searches for and adds medicines, specifying dosage, duration, and instructions for each. 4. The Doctor reviews the complete prescription. 5. The Doctor submits the prescription. 6. The system saves the prescription linked to the consultation. 7. The system automatically sets the prescription status to 'Pending Dispense'. 8. The prescription becomes visible in the Pharmacy module.
Alternative Flows	The searched medicine does not exist in the inventory. The system displays: 'Medicine not found in inventory.' The Doctor adds a free-text note for the Pharmacy Staff

Table 7-18: Use case of View Pending Prescription

Use Case ID	18
Use Case Name	View Pending Prescriptions
Description	Allow pharmacy staff to view prescriptions awaiting dispensing.

Actor	Pharmacy Staff
Pre-Conditions	The Pharmacy Staff is authenticated. Prescriptions with status 'Pending Dispense' exist.
Post Conditions	The Pharmacy Staff has access to all pending prescription details.
Main Flow	1. The Pharmacy Staff navigate to the Pharmacy module. 2. The system displays all prescriptions with status 'Pending Dispense'. 3. The Pharmacy Staff selects a prescription to view details. 4. The system displays patient information, prescribed medicines, dosage, duration, and instructions
Alternative Flows	No pending prescriptions exist. The system displays: 'No prescriptions pending dispensing.'

Table 7-19: Use case of Dispense Medicines

Use Case ID	19
Use Case Name	Dispense Medicines
Description	Handles medicine dispensing, stock deduction, and updating prescription status.
Actor	Pharmacy Staff
Pre-Conditions	The Pharmacy Staff is authenticated. A prescription with status 'Pending Dispense' is selected. Sufficient stock exists in inventory.
Post Conditions	The medicines are dispensed and inventory is updated. The prescription status is updated to 'Fully Dispensed' or 'Partially Dispensed'. The dispensing event is recorded in the audit log
Main Flow	1. The Pharmacy Staff selects a pending prescription. 2. The system displays prescription details and current stock levels for each medicine. 3. The Pharmacy Staff validates the prescription details.

	4. The Pharmacy Staff confirms dispensing of all medicines. 5. The system deducts dispensed quantities from the inventory. 6. The system updates the prescription status to 'Fully Dispensed'. 7. The system records the dispensing action with staff ID and timestamp in the audit log. 8. The system prevents re-dispensing of the completed prescription
Alternative Flows	One or more medicines have insufficient stock. The system highlights the items with low or zero stock. The Pharmacy Staff dispenses available items and marks the prescription as 'Partially Dispensed'. The remaining items are flagged for follow-up.

Table 7-20: Use case of Manage Medicine Records

Use Case ID	20
Use Case Name	Manage Medicine Records
Description	Allow pharmacy staff to add, update, or deactivate medicine records.
Actor	Pharmacy Staff
Pre-Conditions	The Pharmacy Staff is authenticated
Post Conditions	The medicine record is created, updated, or deactivated as requested. The change is recorded in the audit log.
Main Flow	1. The Pharmacy Staff navigate to the Inventory Management module. 2. The Pharmacy Staff selects Add New Medicine. 3. The system displays the medicine creation form. 4. The Pharmacy Staff enters medicine name, category, unit, stock quantity, low-stock threshold, and expiry date. 5. The Pharmacy Staff submits the form.

	6. The system validates the data and creates the medicine record. 7. The medicine is available for prescribing and dispensing.
Alternative Flows	The Pharmacy Staff selects an existing medicine record and clicks Edit. The Pharmacy Staff modifies the required fields. The Pharmacy Staff saves the changes.

Table 7-21: Use case of Track Stock Levels and Expiry

Use Case ID	21
Use Case Name	Track Stock Levels and Expiry
Description	Monitors stock levels, expiry dates, and generate alerts for low stock or expired medicines.
Actor	Pharmacy Staff
Pre-Conditions	The Pharmacy Staff is authenticated. Medicine records exist in the system.
Post Conditions	Inventory reflects accurate stock levels. Expired medicines are marked unavailable. Low stock alerts are generated as appropriate.
Main Flow	1. The Pharmacy Staff navigate to the Inventory module. 2. The system displays a list of all medicines with current stock levels and expiry dates. 3. The system highlights medicines that are below the low-stock threshold. 4. The system flags medicines that are expired or near expiry. 5. The Pharmacy Staff reviews the flagged items. 6. The Pharmacy Staff updates stock quantities as new supplies arrive. 7. The system marks expired medicines as unavailable
Alternative Flows	Stock level for medicine falls below the configured threshold. The system generates low stock alert notification. The Pharmacy Staff is notified to reorder stock.

Table 7-22: Use case of Generate Invoice

Use Case ID	22

Use Case Name	Generate Invoice
Description	Enables Admin to create invoices based on consultation and medicine charges.
Actor	Admin
Pre-Conditions	The admin is authenticated. A consultation has been completed and medicines dispensed.
Post Conditions	The invoice is generated with status 'Pending'. The invoice is retrievable for future reference.
Main Flow	1. The admin navigates to the Billing module. 2. The admin searches for the patient or selects from a completed consultation. 3. The system auto-populates consultation charges and medicine costs. 4. The admin reviews the invoice details. 5. The admin confirms and generates the invoice. 6. The system creates the invoice with status 'Pending'. 7. The system assigns an invoice number and timestamps the record
Alternative Flows	The admin notices an error in the auto-populated charges. The admin edits the applicable line items. The admin confirms the corrected invoice.

Table 7-23: Use case of Record Payment

Use Case ID	23
Use Case Name	Record Payment
Description	Allow Admin to record payments and update invoice status.
Actor	Admin
Pre-Conditions	The admin is authenticated. A 'Pending' invoice exists for the patient.
Post Conditions	The payment is recorded and the invoice status is updated to 'Paid'. The payment is logged in the audit log.
Main Flow	1. The admin opens the pending invoice.

	2. The system displays the invoice total and line items. 3. The admin confirms that cash payment has been received. 4. The admin records the payment. 5. The system updates the invoice status to 'Paid'. 6. The system records the payment with the date, amount, and Admin ID. 7. The system logs the payment event in the audit log
Alternative Flows	The patient is unable to pay the full amount. The admin records the amount paid. The system keeps the invoice status as 'Pending' with a partial amount noted. The remaining balance is tracked for future collection. The admin searches for a prior invoice by patient name, invoice number, or date. The system retrieves and displays the historical invoice. The admin views or reprints the invoice as required.

Table 7-24: Use case of Generate Clinic Reports

Use Case ID	24
Use Case Name	Generate Clinic Reports
Description	Provides reports on clinic operations such as patient visits and consultations.
Actor	Super Admin, Admin
Pre-Conditions	The user is authenticated with the appropriate role. Historical appointment and consultation data exists.
Post Conditions	The clinic report is generated and displayed. The report can be exported for offline use.
Main Flow	1. The user navigates to the Reports module. 2. The user selects the Clinic Reports category. 3. The user specifies the report parameters (date range, doctor, report type). 4. The system compiles and generates the report.

	5. The system displays the report including patient counts, consultation counts, and queue statistics. 6. The user can export the report in a supported format.
Alternative Flows	Report generation exceeds the acceptable time threshold. The system displays a timeout notification. The user is prompted to reduce the date range and regenerate

Table 7-25: Use case of Generate Pharmacy Reports

Use Case ID	25
Use Case Name	Generate Pharmacy Reports
Description	Generate reports related to medicine usage and inventory.
Actor	Super Admin, Pharmacy staff
Pre-Conditions	The user is authenticated. Dispensing and inventory data exists.
Post Conditions	The pharmacy report is generated and available for review or export.
Main Flow	1. The user navigates to the Pharmacy Reports section. 2. The user selects a type of report (stock usage, expiry report, dispensing history). 3. The user specifies the date range or other filter parameters. 4. The system compiles and displays the pharmacy report. 5. The user can export the report in a supported format.

Table 7-26: Use case of View Audit Logs

Use Case ID	26
Use Case Name	View Audit Logs
Description	Allows Super Admin to monitor system activities for security and compliance.
Actor	Super Admin
Pre-Conditions	The Super Admin is authenticated. Audit log data exists.

Post Conditions	The Super Admin has reviewed the required audit log entries. Exported audit logs are available for compliance or investigation purposes.
Main Flow	1. The Super Admin navigates to the Audit Log section. 2. The system displays recent audit log entries. 3. The Super Admin applies filters (date range, user, action type, module). 4. The system returns filtered audit log entries. 5. The Super Admin reviews individual log entries for details. 6. The Super Admin can export the filtered audit log.
Alternative Flows	No log entries match the applied filters. The system displays: 'No audit log entries found for the selected criteria.' The Super Admin adjusts the filters.

Table 7-27: Use case of Book Appointment Without login

Use Case ID	27
Use Case Name	Book Appointment Without Login
Description	Allows patients to book appointments without creating an account
Actor	Patient
Pre-Conditions	The clinic self-service portal is accessible. Doctor schedules are configured and available.
Post Conditions	The appointment is created with status 'Scheduled'. The patient receives confirmation.
Main Flow	1. The patient navigates to the clinic's self-service booking page. 2. The system displays the appointment booking form. 3. The patient enters personal details (name, phone number, and date of birth). 4. The patient selects the preferred doctor, date, and available time slot.

	5. The patient submits the booking request. 6. The system validates the details and checks slot availability. 7. The system creates the appointment with status 'Scheduled'. 8. The system sends a booking confirmation notification to the patient
Alternative Flows	The selected slot is taken at the time of submission. The system displays available alternative slots. The patient selects an alternative slot and resubmits.

Table 7-28: Use case of Interact with Clinic Chatbot

Use Case ID	28
Use Case Name	Interact with Clinic Chatbot
Description	Provides automated responses to patient queries using a knowledge base.
Actor	Patient
Pre-Conditions	The chatbot interface is accessible on the clinic website or portal. Approved knowledge source documents are uploaded and indexed
Post Conditions	The patient receives accurate, approved information. No medical diagnosis or prescription advice is provided by the chatbot.
Main Flow	1. The patient opens the chatbot interface. 2. The chatbot greets the patient and presents query options. 3. The patient types a query (e.g., clinic hours, doctor's availability, health information). 4. The system processes the query against the predefined approved knowledge base. 5. The chatbot displays a relevant response to the patient. 6. The patient may ask follow-up questions within the allowed scope.

Alternative Flows	The patient asks for a diagnosis, prescription recommendation, or information not in the knowledge base. The chatbot responds: 'I'm sorry, I am unable to provide medical diagnoses or prescriptions. Please consult a doctor.' The chatbot offers help with other approved topics.

Table 7-29: Use case of Manage Chatbot Knowledge Base

Use Case ID	29
Use Case Name	Manage Chatbot Knowledge Base
Description	Allows authorized users to update chatbot information sources.
Actor	Doctor
Pre-Conditions	The authorized user is authenticated. The Knowledge Base Management section is accessible.
Post Conditions	The chatbot knowledge base is updated with approved information. The upload or removal event is logged in the audit log.
Main Flow	1. The Doctor or Super Admin navigates to the Chatbot Knowledge Base section. 2. The system displays a list of currently uploaded knowledge source documents. 3. The user selects Upload New Document. 4. The user selects an approved health-related PDF document. 5. The system uploads and indexes the document for chatbot use. 6. The system confirms successful upload and indexing. 7. The chatbot knowledge base is updated to include the new document.
Alternative Flows	The user selects an existing document and clicks Remove. The system prompts confirmation. Upon confirmation, the document is removed from the knowledge base. The chatbot no longer references the removed document.

Table 7-30: Use case of Mark Prescription for External Dispensing

Use Case ID	30
Use Case Name	Mark Prescription for External Dispensing
Description	Allow pharmacy staff to mark prescriptions for off-site delivery.
Actor	Pharmacy Staff
Pre-Conditions	The Pharmacy Staff is authenticated. A prescription with status 'Pending Dispense' exists.
Post Conditions	The prescription is flagged for external dispensing. The Pharmacy Staff proceeds with identity verification.
Main Flow	1. The Pharmacy Staff opens a pending prescription. 2. The Pharmacy Staff selects the option Mark for External Dispensing. 3. The system displays the external dispensing form. 4. The Pharmacy Staff confirms the patient's off-site status. 5. The system flags the prescription for external dispensing processing. 6. The system proceeds with patient identity verification.
Alternative Flows	Prescription Not Eligible: The prescription has already been dispensed or cancelled. The system prevents action and displays an appropriate status message

Table 7-31: Use case of Verify Patient Identity for External Dispensing

Use Case ID	31
Use Case Name	Verify Patient Identity for External Dispensing
Description	Ensures correct patient identity before external medicine delivery.
Actor	Pharmacy Staff
Pre-Conditions	The prescription is flagged for external dispensing. The Pharmacy Staff is authenticated.
Post Conditions	Patient identity is verified. The system allows the dispensing step to proceed.
Main Flow	1. The system displays the patient identity verification form.

	2. The Pharmacy Staff selects the verification method: phone number, clinic ID, or QR code scan. 3. The Pharmacy Staff enters or scans the identity information. 4. The system validates the provided identity against the patient record. 5. Upon successful verification, the system proceeds to the dispensing step.
Alternative Flows	Identity Verification Failure: The provided identity does not match the patient record. The system displays: 'Identity verification failed. Please verify and try again.' The Pharmacy Staff may attempt an alternative verification method. After repeated failures, the dispensing process is cancelled and logged.

Table 7-32: Use case of Complete External Dispensing

Use Case ID	32
Use Case Name	Complete External Dispensing
Description	Finalizes external dispensing, updates inventory, and records delivery details.
Actor	Pharmacy staff
Pre-Conditions	The Pharmacy Staff is authenticated. Patient identity verification is completed. Sufficient inventory stock exists.
Post Conditions	Inventory is updated immediately. Prescription status is set to 'Dispensed (External)'. A delivery record is generated and stored. The event is recorded in the audit log for compliance.
Main Flow	1. The system displays prescription details for external dispensing. 2. The Pharmacy Staff confirms the medicines to be dispensed. 3. The Pharmacy Staff selects the payment method.

	4. The Pharmacy Staff confirms the dispensing action. 5. The system deducts dispensed quantities from the inventory immediately. 6. The system updates the prescription status to 'Dispensed (External)'. 7. The system generates a delivery record including patient ID, staff ID, date/time, and payment method. 8. The system logs the external dispensing event in the audit log
Alternative Flows	Insufficient Stock: One or more medicines are out of stock. The system highlights the unavailable items. The Pharmacy Staff dispenses available items and notes partial fulfilment. The remaining items are flagged for follow-up.

CHAPTER 8: ACTIVITY DIAGRAMS

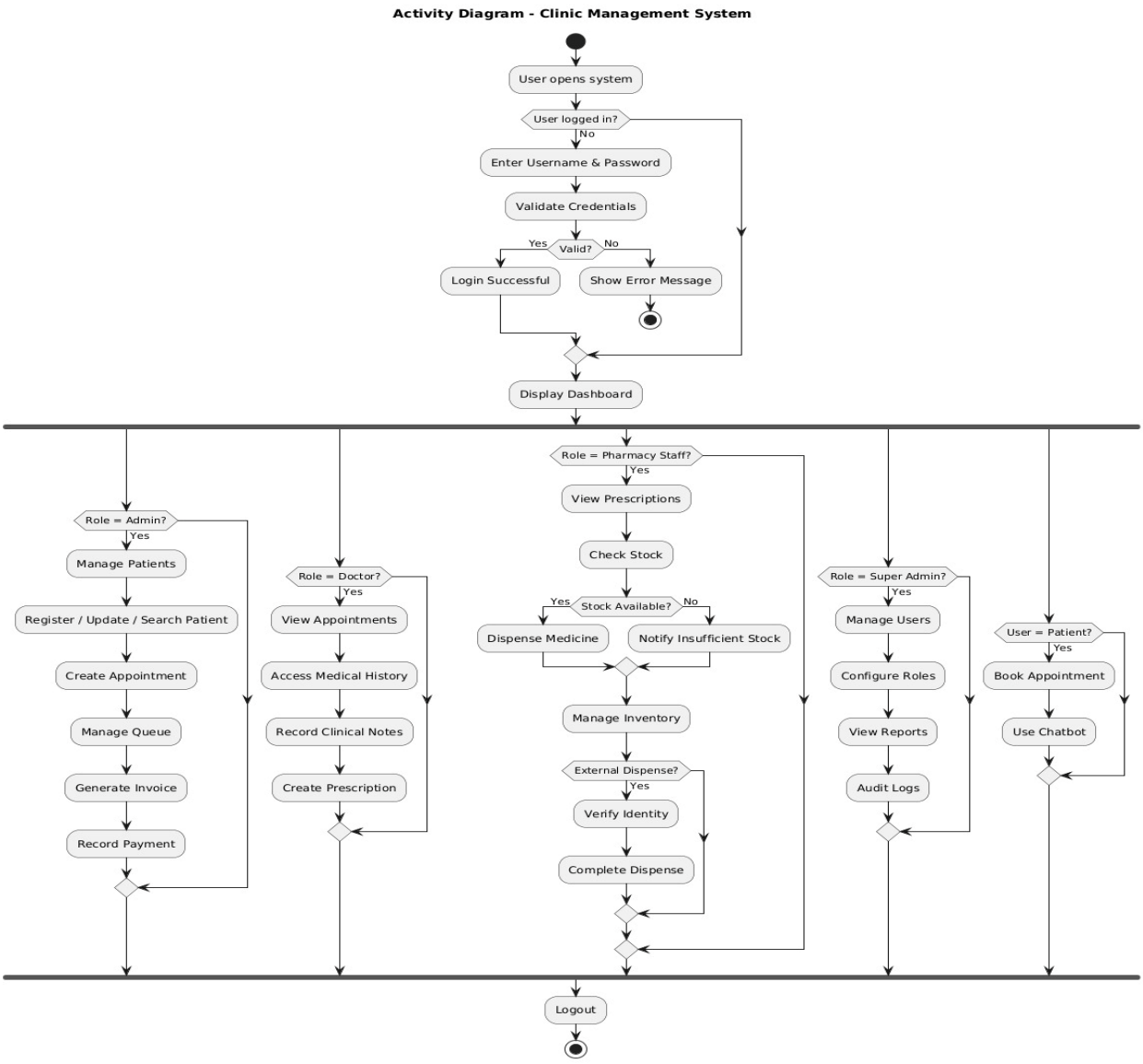


Figure 8-1: Activity Diagram - Overall System

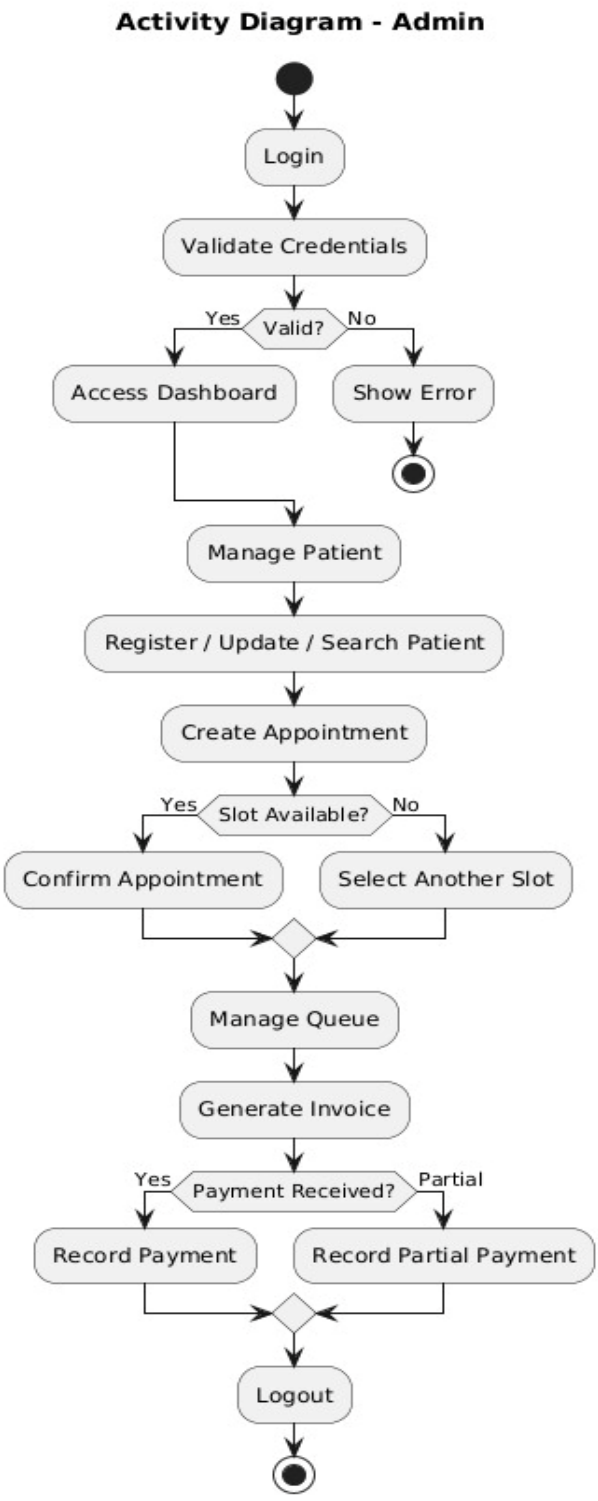


Figure 8-2: Activity Diagram - Admin

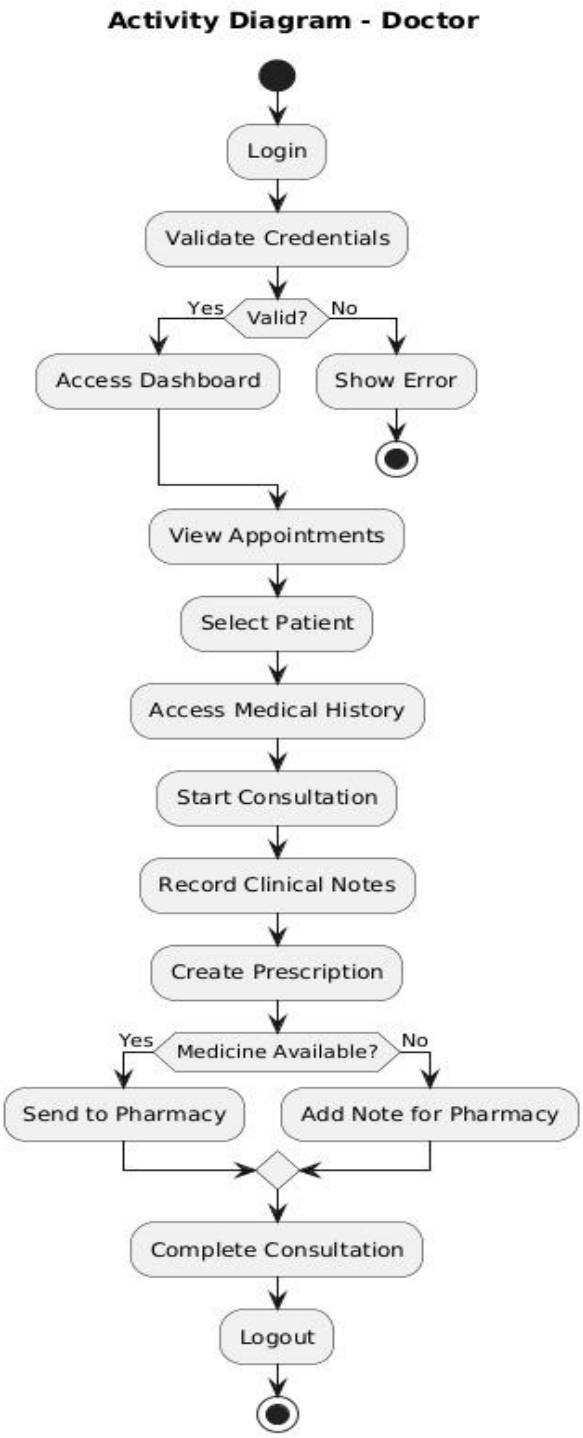


Figure 8-3: Activity Diagram - Doctor

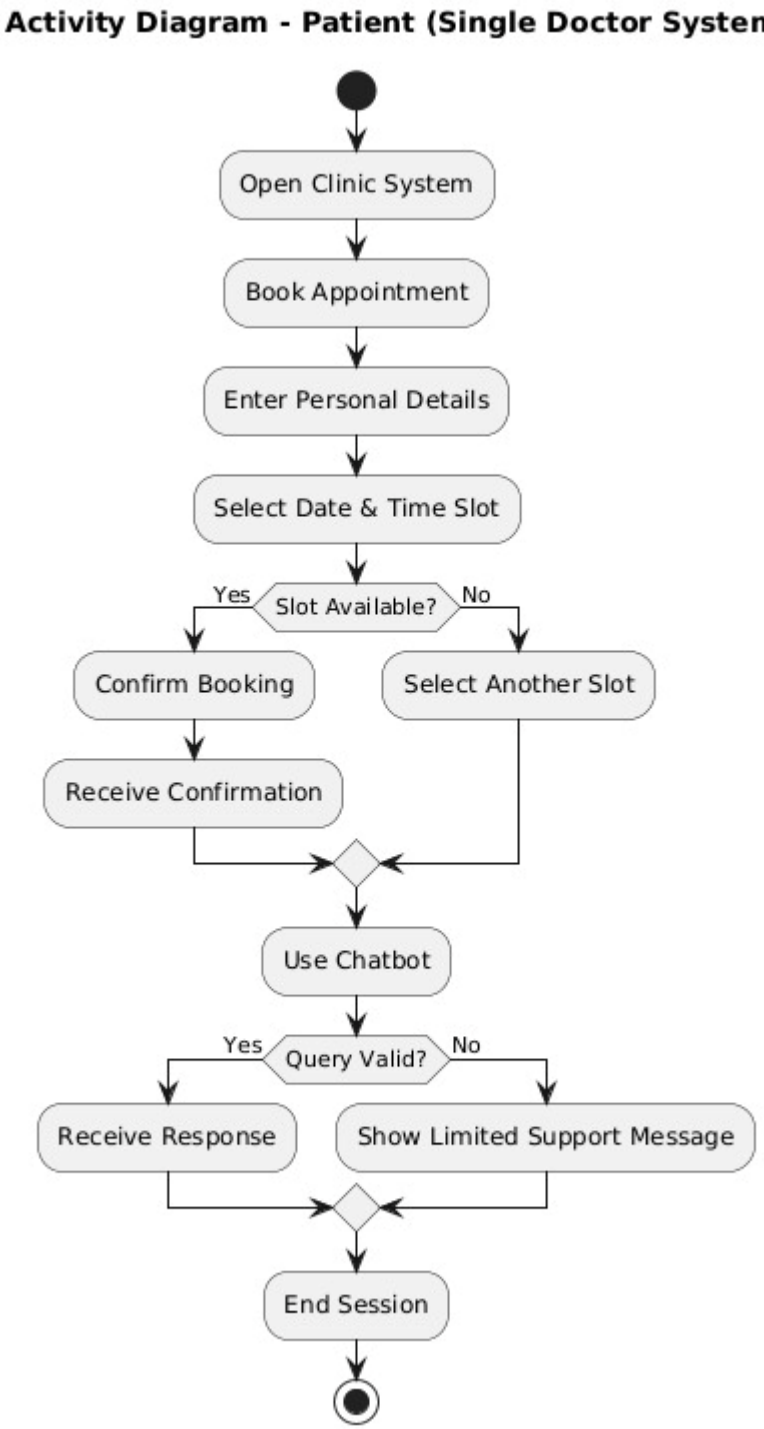


Figure 8-4: Activity Diagram - Patient

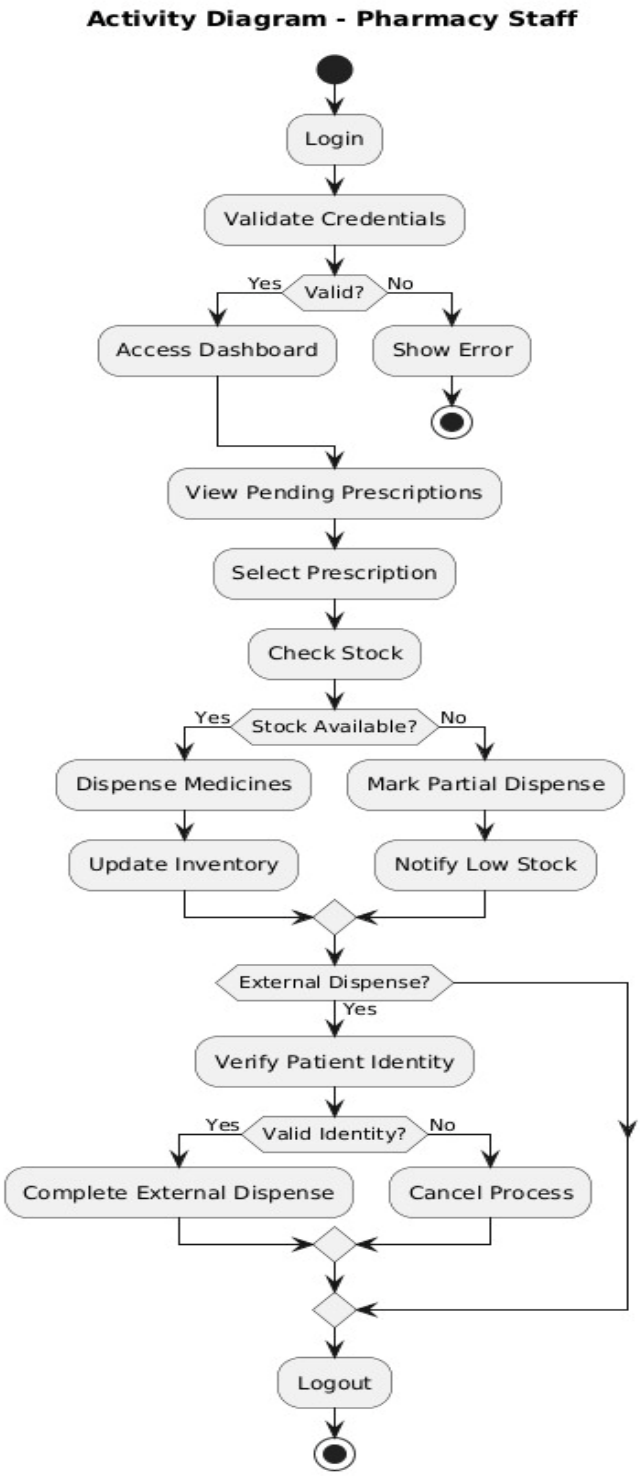


Figure 8-5: Activity Diagram - Pharmacy Staff

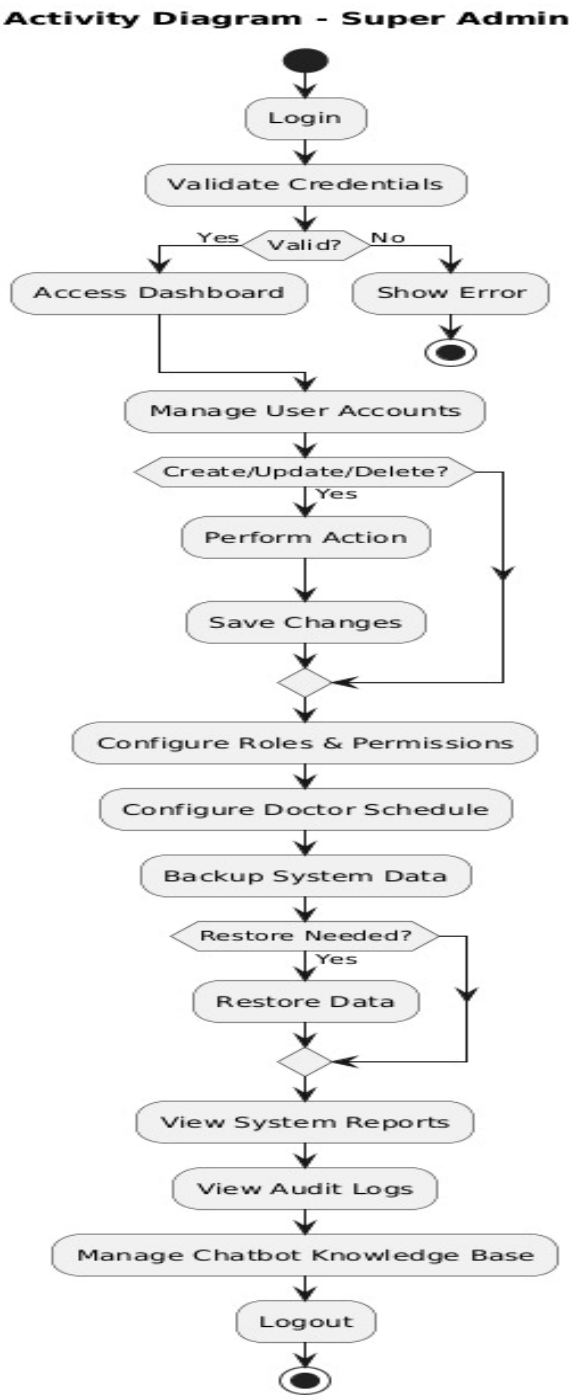


Figure 8-6: Activity Diagram - Super Admin

CHAPTER 9: EXTERNAL INTERFACE REQUIREMENTS

9.1 User Interface Requirements

A role-based, responsive web interface shall be provided by the system, designed specifically for clinic and pharmacy workflows.

General UI Requirements

- Separate dashboards shall be provided for:
    - Doctor (patient records, prescriptions)
    - Pharmacist (inventory, dispensing, alerts)
    - Super Admin (user management, reports)
- The interface shall be responsive and accessible on desktops and tablets.
- Consistent layouts and navigation menus shall be used across all modules.

Clinic-Specific UI Features

- A patient registration form shall be provided, including:
    - Personal details
    - Medical history
- An appointment booking interface shall be provided, including:
    - Date and time slot selection
    - Availability display
- The doctor dashboard shall display:
    - Daily appointments
    - Patient history
    - Prescription creation panel

Pharmacy-Specific UI Features

- An inventory dashboard shall be provided, displaying:
    - Medicine name, quantity, batch number, expiry date
- Visual indicators shall be provided for:
    - Low stock medicines
    - Near-expiry medicines
- A prescription view screen shall be provided for pharmacists to:
    - Verify doctor prescriptions

- Dispense medicines

Alert & Notification UI

- Pop-up and dashboard alerts shall be provided for:
  - Stock alerts
  - Expiry alerts
- Appointment reminders shall be provided through a notification panel accessible to all roles.

Accessibility Requirements

- Readable fonts and proper color contrast shall be ensured
- The UI shall be simple for non-technical users.
- Minimal steps shall be required to complete key tasks (e.g., booking, dispensing).

9.2 Hardware Interface Requirements

- The system is a web-based application and does not require specialized hardware.
  - The following hardware dependencies shall apply:
    - A server or cloud hosting environment shall be provided with at least:
      - 2 vCPU
      - 4 GB RAM
      - 50 GB storage
    - Clinic staff devices (computers or tablets) shall:
      - Support a modern web browser
      - Have stable internet connectivity
    - A printer (optional) may be used for:
      - Printing prescriptions
      - Printing reports
- No direct integration with specialized medical devices shall be required in this version

9.3 Software Interface Requirements

Integration with the following software components shall be provided:

- Database System:

  A relational database management system (e.g., MySQL) shall be used by the system for storing:
  - Patient records
  - Appointment details
  - Prescription data
  - Medicine inventory
- Email Service:

  Integration with an email service (SMTP) shall be provided by the system for:
  - Password reset notifications
  - Sending patient bills
- SMS Notification Service:

  Appointment confirmation messages shall be sent by the system.
- Chatbot Module:

  A chatbot interface shall be included in the system to:
  - Handle patient queries
  - Provide appointment guidance
  - Respond to frequently asked questions
- Web Browser Compatibility:

  The system shall be made accessible through modern web browsers such as:
  - Google Chrome
  - Microsoft Edge
  - Mozilla Firefox
- Application Server:

  Deployment shall be carried out on a web application server compatible with the backend framework used.

9.4 Communication Interface Requirements

- All client-server communications shall be conducted over HTTPS using TLS 1.2 or higher to ensure secure data transmission.
- RESTful APIs shall be used for communication between frontend and backend components.
- Data exchanged between components shall be in JSON format.

- Operation shall occur over standard TCP/IP networks; no proprietary communication protocols shall be required.
- Real-time updates shall be supported for:
    - Appointment status
    - Stock and expiry alerts
- All communication involving sensitive data (e.g., patient records) shall be encrypted and securely handled.

CHAPTER 10: DATA REQUIREMENTS

This chapter defines the data entities, their attributes, relationships, and storage requirements for the OneCare Clinic Integrated Pharmacy Management System.

10.1 Data Entities and Attributes

10.1.1 Patient

All personal and medical information for registered patients is stored.

Table 10-1: Patient Entity Specification

Field	Data Type	Constraints	Description
patient_id	VARCHAR (10)	PRIMARY KEY, UNIQUE, NOT NULL	Unique patient identifier
first_name	VARCHAR (100)	NOT NULL	Patient's first name
last_name	VARCHAR (100)	NOT NULL	Patient's last name
date_of_birth	DATE	NOT NULL	Patient date of birth
gender	ENUM('Male','Female','Other')	Male/Female/Other	Patient gender
contact_number	VARCHAR (15)	NOT NULL	Primary contact number
email	VARCHAR (150)	UNIQUE	Email address
address	TEXT	NULLABLE	Residential address
blood_group	VARCHAR (5)	NULLABLE	Patient blood group
registered_date	DATETIME	NOT NULL, DEFAULT NOW ()	Date of registration

10.1.2 User (System Users)

Stores credentials and role assignments for all system users including Doctors, Pharmacists, and Admins.

Table 10-2: User Entity Specification

Field	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user identifier
username	VARCHAR (50)	UNIQUE, NOT NULL	Login username
password_hash	VARCHAR (255)	NOT NULL	Bcrypt/Argon2 hashed password
role	ENUM ('Doctor','Pharmacist','Admin','Super Admin')	Doctor/Pharmacist/Admin/Super Admin	User role for RBAC
email	VARCHAR (150)	UNIQUE, NOT NULL	Email for notifications/2FA
is_active	BOOLEAN	DEFAULT TRUE	Account active status
failed_attempts	INT	DEFAULT 0	Consecutive failed logins
locked_until	DATETIME	NULLABLE	Account lockout expiry
last_login	DATETIME	NULLABLE	Timestamp of last login

10.1.3 Appointment

Manages all appointment booking data between patients and the doctor.

Table 10-3: Appointment Entity Specification

Field	Data Type	Constraints	Description
appointment_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique appointment ID
patient_id	VARCHAR (10)	FOREIGN KEY → Patient	Associated patient
appointment_date	DATE	NOT NULL	Date of appointment
appointment_time	TIME	NOT NULL	Scheduled time slot

status	ENUM (`Scheduled`,`Cancelled`,`Completed`,`No-show`)	Pending/Confirmed/Cancelled/Completed	Appointment status
reason	TEXT	NULLABLE	Reason for visit
created_at	DATETIME	DEFAULT NOW ()	Booking timestamp

10.1.4 Prescription

Records digital prescriptions issued by the doctor to patients.

Table 10-4: Prescription Entity Specification

Field	Data Type	Constraints	Description
prescription_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique prescription ID
patient_id	VARCHAR (10)	FOREIGN KEY → Patient	Associated patient
doctor_id	INT	FOREIGN KEY → User	Issuing doctor
appointment_id	INT	FOREIGN KEY → Appointment, NULLABLE	Linked appointment
clinical_notes	TEXT	NULLABLE	Doctor's consultation notes
issued_date	DATETIME	NOT NULL, DEFAULT NOW ()	Date prescription was created
status	ENUM ('Issued','Dispensed','Cancelled')	Issued/Dispensed/Cancelled	Prescription fulfilment status

10.1.5 Prescription Item

Stores individual medicine line items linked to a prescription.

Table 10-5: Prescription Item Entity Specification

Field	Data Type	Constraints	Description
item_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique item identifier

prescription_id	INT	FOREIGN KEY → Prescription	Parent prescription
medicine_id	INT	FOREIGN KEY → Medicine	Prescribed medicine
dosage	VARCHAR (100)	NOT NULL	Dosage instruction (e.g. 500mg)
frequency	VARCHAR (100)	NOT NULL	Intake frequency (e.g. twice daily)
duration_days	INT	NOT NULL	Treatment duration in days
quantity	INT	NOT NULL	Total quantity to dispense

10.1.6 Medicine (Inventory)

Maintains the pharmacy's medicine catalogue and stock levels.

Table 10-6: Medicine (Inventory) Entity Specification

Field	Data Type	Constraints	Description
medicine_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique medicine identifier
name	VARCHAR (150)	NOT NULL	Medicine name
generic_name	VARCHAR (150)	NULLABLE	Generic / INN name
category	VARCHAR (100)	NOT NULL	Drug category
unit	VARCHAR (50)	NOT NULL	Unit of measure (tablet/ml/etc.)
quantity_in_stock	INT	NOT NULL, DEFAULT 0	Current available stock
reorder_level	INT	NOT NULL	Minimum stock before alert
expiry_date	DATE	NOT NULL	Medicine expiry date
is_quarantined	BOOLEAN	DEFAULT FALSE	Expired/quarantined flag
unit_price	DECIMAL (10,2)	NOT NULL	Price per unit

10.1.7 Audit Log

Records all system user actions for security and compliance purposes.

Table 10-7: Audit Log Entity Specification

Field	Data Type	Constraints	Description
log_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique log entry identifier
user_id	INT	FOREIGN KEY → User	User who performed the action
action_type	VARCHAR (100)	NOT NULL	Type of action performed
entity_type	VARCHAR (100)	NULLABLE	Affected data entity
entity_id	VARCHAR (50)	NULLABLE	ID of affected record
description	TEXT	NULLABLE	Detailed action description
ip_address	VARCHAR (45)	NULLABLE	Client IP address
timestamp	DATETIME	NOT NULL, DEFAULT NOW ()	Timestamp of the action

10.1.8 External Dispensing

Tracks prescriptions fulfilled outside the clinic's pharmacy

Table 10-8: External Dispensing Entity Specification

Field	Data Type	Constraints	Description
external_dispense_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for external dispensing
prescription_id	INT	FOREIGN KEY → Prescription	Prescription being dispensed externally
patient_id	VARCHAR (10)	FOREIGN KEY → Patient	Patient receiving the external dispense

staff_id	INT	FOREIGN KEY → User	pharmacist performing the external dispense
dispense_date	DATETIME	NOT NULL, DEFAULT NOW ()	Date and time of external dispense
quantity_dispensed	INT	NOT NULL	Total quantity of medicine dispensed
status	ENUM('Dispensed','Partially Dispensed')	NOT NULL	Fulfilment status
delivery_method	ENUM('Pick-up','Delivery')	NOT NULL	How medicine is delivered
payment_method	ENUM('Cash')	NOT NULL	Payment method (aligns with BR-011)
audit_log_id	INT	FOREIGN KEY → Audit Log	Link to audit trail for compliance

10.2 Data Relationships

The key relationships between data entities are as follows:

- A Patient can have many Appointments (one-to-many).
- An Appointment can have one Prescription (one-to-one).
- A Prescription can have many Prescription Items (one-to-many).
- A Prescription Item references one Medicine from the inventory (many-to-one).
- A User (Doctor) can issue many Prescriptions (one-to-many).
- All User actions are recorded in the Audit Log (one-to-many).
- A Prescription can have zero or one External Dispense (optional one-to-one).
- An External Dispense references one Patient.
- External Dispense is performed by one User (Pharmacist).
- External Dispense updates one or more Medicine stock levels in Inventory.
- All External Dispense actions are recorded in the Audit Log.

10.3 Data Storage and Retention

- All data shall be stored in a MySQL 8.0 relational database deployed via Docker.
- Patient medical records shall be retained for a minimum of 7 years in accordance with applicable health data retention policies.
- Audit log records shall be retained for a minimum of 90 days in a centralized logging system.
- Automated database backups shall be performed every 24 hours and stored in secure cloud storage.
- Expired medicine records shall be flagged as quarantined but not deleted, to maintain historical dispensing traceability.
- External dispensing records shall be retained for a minimum of 7 years along with Prescription and Patient data.
- Inventory updates due to external dispensing shall maintain historical records for auditing purposes.
- Delivery and payment details for external dispensing shall be stored in the database and linked to the corresponding Prescription.

10.4 Data Integrity and Validation Rules

- Patient IDs must be unique and system-generated at registration.
- All prescription items must reference a valid, non-quarantined medicine in the inventory.
- Appointment dates must not be set in the past.
- Stock quantity must never fall below zero; negative stock updates shall be rejected by the system.
- All monetary values (unit prices, invoices) shall be stored as DECIMAL (10,2) to prevent floating point errors.
- Foreign key constraints shall be enforced at the database level to prevent orphaned records.
- External Dispense quantity must not exceed the remaining Prescription quantity.
- Patient identity must be verified (phone number, clinic ID, or QR code) before external dispensing.
- External Dispense actions must create corresponding Audit Log entries.

- Stock updates from external dispensing must enforce non-negative inventory; insufficient stock shall block the dispense.
- Prescription status shall reflect partial or full external dispensing.

CHAPTER 11: ALTERNATIVE SOLUTIONS CONSIDERED

Table 11-1: Analysis of Alternative Solutions and Reasons for Rejection

Alternative	**Description**	**Reason for Rejection**
Manual / Paper-Based System	Continue managing patient records, appointments, and pharmacy stock using paper files and registers.	Prone to human errors; difficult to maintain records; no real-time updates; poor inventory tracking; time-consuming.
Phone-Based Appointment Booking	Patients book appointments via phone calls without any digital system.	No proper scheduling system: high chance of missed or overlapping appointments; no record tracking; inefficient communication.
Separate Clinic and Pharmacy Systems	Use two different systems for patient management and pharmacy operations.	Lack of integration; data duplication; inefficient workflow between doctor and pharmacist; increased operational complexity.
Spreadsheet-Based System	Use tools like spreadsheets to manage patient data and medicine inventory.	Not suitable for multi-user access; lacks automation; no real-time alerts (stock/expiry); high risk of data inconsistency.
Mobile Application Only	Develop only a native mobile app instead of a web-based system.	Higher development cost; requires separate apps for different platforms; less accessible for clinic staff compared to web system.
Manual + Partial Digital System	Use a combination of manual records and basic digital tools.	Data inconsistency; duplication of work; increased chances of errors; inefficient workflow.
Admin Approval for Appointment Booking	After a patient has booked an appointment, the administrator must manually review and approve each	Causes delays in appointment confirmation; increases admin workload; inefficient for small clinics with limited staff; real-time

	booking before confirmation.	booking without approval is faster and more user-friendly.
Online Payment for Appointment Booking	Require patients to make online payments (card or digital payment) when booking appointments.	Not suitable for rural areas where most patients do not use online payments; may reduce accessibility; cash-based payment at the clinic is more practical and convenient.

CHAPTER 12: FEASIBILITY STUDY

This chapter assesses the feasibility of the OneCare Clinic Integrated Pharmacy Management System across four dimensions: technical, economic, operational, and schedule feasibility.

12.1 Technical Feasibility

Technical feasibility evaluates whether the proposed system can be built using available technologies, tools, and the development team's existing skill sets.

12.1.1 Technology Stack Assessment

Table 12-1: Proposed Technology Stack and Justification

Component	**Proposed Technology**	**Justification**
Frontend	React.js	Component-based, responsive design, widely supported
Backend	Spring Boot	Scalable, modular, well-documented, suitable for REST APIs.
Database	MySQL 8.0	Relational model suitable for clinic records; Docker-compatible
Authentication	JWT + bcrypt/Argon2	Stateless secure authentication; industry standard.
Containerization	Docker	Consistent deployment environment, OS-independent.
Notifications	SMS API (e.g. e-SMS or Twillio)	Reliable SMS delivery for appointment reminders
API Documentation	Swagger / OpenAPI	Standardized, auto-generated API documentation

12.1.2 Technical Feasibility Assessment

- The development team has prior experience with the proposed technology stack through coursework and projects.

- All tools and frameworks are open-source, eliminating licensing costs.
- Modular system architecture separates frontend, backend, and database layers.
- Docker ensures environment consistency across clinic devices.
- Offline operations, including queue updates, appointment logging, and external dispensing, introduce moderate technical complexity but are achievable via local caching and service worker strategies.
- External dispensing functionality interacts with inventory, patient verification, and audit logging; the system design supports these workflows without introducing new dependencies.

Conclusion: The project is technically feasible. The chosen technologies are mature, widely documented, and well within the competence of the development team.

12.2 Economic Feasibility

Economic feasibility examines whether the system can be developed and operated within acceptable financial boundaries relative to the benefits it provides.

12.2.1 Cost Estimate

Table 12-2: Cost Estimation

Cost Item	Estimated Cost (LKR)
Development (student project - no labour cost)	0
Cloud hosting (e.g. AWS / DigitalOcean – 12 months)	~60,000 – 120,000
Domain name registration (1 year)	~3,000 – 5,000
SSL certificate (Let's Encrypt – free)	0
SMS notification service (Twilio / local provider)	~10,000 – 20,000 per year
Software licenceslicenses (open-source stack)	0
Total Estimated Annual Operational Cost	~73,000 – 145,000 LKR

12.2.2 Benefit Analysis

- Elimination of paper-based record management reduces stationery and storage costs.

- Reduction in appointment management errors decreases patient no-shows, improving revenue consistency.
- Automated inventory alerts prevent stock-outs and reduce losses from expired medicines.
- Digital prescriptions eliminate re-dispensing errors, reducing potential liability.
- Estimated productivity gain of 30–40% in administrative tasks based on similar clinic system deployments.

Conclusion: The system is economically feasible. The operational costs are modest relative to the efficiency gains and error reduction benefits for a small clinic environment.

12.3 Operational Feasibility

Operational feasibility assesses whether the proposed system will be accepted and successfully used by the clinic's staff within the day-to-day working environment.

12.3.1 User Readiness

- Clinic staff are assumed to have basic computer literacy (ASM-001).
- The system is designed for ease of learning: a new staff member shall be able to perform basic tasks within 30 minutes of first use (NFR-U002).
- All critical tasks require no more than three navigation steps (NFR-U003), reducing the learning curve.

12.3.2 Organizational Impact

- The transition from manual to digital workflows will require a one-time data migration effort for existing patient records and medicine inventory.
- An initial training session for all staff roles (Doctor, Pharmacist, Admin) is planned prior to system go-live.
- The system does not require changes to the clinic's physical infrastructure beyond provision of computer workstations, which are already available.
- The Super Admin role ensures ongoing user management without dependence on external technical support for routine tasks.

Conclusion: The system is operationally feasible. The intuitive interface design and multilingual support directly address the identified operational barriers.

12.4 Schedule Feasibility

Schedule feasibility evaluates whether the system can be delivered within the planned timeline.

12.4.1 Estimated Project Timeline

Table 12-3: Estimated Project Timeline Description

Phase	Description	Duration	Milestone
1	Requirements Analysis & SRS Documentation	3 weeks	SRS approved
2	System Design (Architecture, Database, UI Mockups)	3 weeks	Design document
3	Frontend Development (React.js, core modules)	6 weeks	UI prototype
4	Backend Development (APIs, Auth, DB integration)	6 weeks	API tested
5	Integration Testing & Bug Fixing	3 weeks	Test reports
6	User Acceptance Testing (UAT) with clinic staff	2 weeks	UAT sign-off
7	Deployment, Data Migration & Training	2 weeks	System go-live
8	Buffer Time	3 weeks	Contingency
Total		~28 weeks	

12.4.2 Schedule Risk Factors

- Four-member team enables parallel development of frontend and backend.
- Academic examination periods may cause schedule interruptions and have been accounted for in the timeline estimates.
- Modular architecture allows phased delivery, mitigating full-system delays.
- Scope has been clearly bounded (excluding online payments, multi-clinic support, and telemedicine) to prevent scope creep.

Conclusion: The schedule is feasible. The 28-week timeline is achievable for a four-member team given the defined scope, provided that parallel development tracks are maintained and scope is strictly controlled.

CHAPTER 13: RISK ANALYSIS

This section identifies and evaluates potential risks associated with the development of the OneCare Clinic Integrated Pharmacy Management System. The analysis focuses on technical, resource, schedule, requirements, and external risks that may impact project execution and delivery. Appropriate mitigation strategies have been defined to minimize the likelihood and impact of each identified risk.

Table 13-1: Risk Analysis Description

Risk ID	Description	Category	Likelihood	Impact	Risk Level	Mitigation Strategy
R1	Unequal distribution of tasks among team members	Resource	High	Medium	High	Tasks shall be clearly defined and allocated based on individual competencies, and progress shall be monitored regularly.
R2	Ineffective communication among team members	Resource	Medium	High	High	Regular team meetings shall be conducted, and collaboration tools (e.g., messaging platforms, task boards) shall be used to ensure continuous communication.
R3	Insufficient technical	Technical	Medium	High	High	Required technical

	expertise in specific development areas					knowledge shall be acquired through documentation, training resources, and guidance from experienced individuals.
R4	Integration issues between frontend and backend components	Technical	Medium	High	High	API contracts shall be defined early, and integration testing shall be performed incrementally throughout development.
R5	High occurrence of software defects during development	Technical	High	Medium	High	Code reviews, unit testing, and continuous debugging practices shall be implemented throughout the development lifecycle.
R6	Delays due to academic workload and internship	Schedule	High	Medium	High	A realistic project schedule shall be prepared, and tasks shall be prioritized and

	commitments					distributed to balance workload effectively.
R7	Scope creep due to uncontrolled addition of new features	Requirements	Medium	High	High	All requirements shall be strictly managed according to the approved SRS, and any changes shall follow a formal change control process.
R8	Misunderstanding or ambiguity in system requirements	Requirements	Medium	High	High	Requirements shall be reviewed regularly with stakeholders, and clarifications shall be documented and approved.
R9	Version control conflicts and improper code management	Technical	Medium	Medium	Medium	A standardized branching strategy shall be followed, and team members shall be trained in version control best practices.

R10	Dependency on a single team member for critical tasks (key person risk)	Resource	Medium	High	High	Knowledge sharing shall be encouraged, and critical tasks shall be documented and distributed among multiple team members.
R11	Data loss during development	Technical	Low	High	Medium	Regular backups shall be maintained, and version control systems shall be used to preserve code and data integrity.
R12	Deployment and environment configuration issues	Technical	Medium	Medium	Medium	Deployment shall be tested in a staging environment prior to production release, and configuration steps shall be documented.
R13	Inaccurate estimation of task duration	Schedule	High	Medium	High	Tasks shall be broken down into smaller units, and progress shall be tracked against

						defined milestones.
R14	Network connectivity issues affecting development activities	External	Medium	Medium	Medium	Alternative internet sources (e.g., mobile hotspots) shall be used, and offline work shall be planned where possible.
R15	Data security and patient privacy risks	Technical	Medium	High	High	Security measures such as authentication, authorization, and data encryption shall be implemented, following standard security practices.
R16	System downtime or performance degradation	Technical	Medium	High	High	System performance shall be monitored, and load handling strategies shall be implemented to ensure availability.
R17	Failure of third-party services	External	Medium	Medium	Medium	Reliable service providers shall be selected, and

	(e.g., SMS, APIs)					fallback mechanisms or alternatives shall be considered.
R18	Resistance from users in adopting the new system	Operational	Medium	Medium	Medium	User training and system demonstrations shall be conducted to improve usability and acceptance.

CHAPTER 14: REFERENCES

[1] ISO25000, “ISO/IEC 25010,” iso25000.com, 2022. https://iso25000.com/index.php/en/iso-25000-standards/iso-25010 (accessed Mar. 09, 2026).

[2] G. D. Almeida, “ChatGPT content included 1. Introduction Sri Lanka’s healthcare sector is undergoing rapid digital transformation, with electronic health records (EHR), telemedicine, mobile health apps, and connected hospital systems becoming integral to patient care.,” *Linkedin.com*, Sep. 16, 2025. https://www.linkedin.com/pulse/compliance-security-framework-sri-lankan-healthcare-de-almeida-w4ikc/ (accessed Mar. 11, 2026).

CLIENT REQUIREMENT SIGN-OFF

The client and project supervisor must review and approve the scope, requirements, and features documented in this SRS before the project proceeds to the design phase. Any changes to the approved requirements after this sign-off must be submitted as a formal Change Request and re-approved by both parties.

14.1 Client Representative Sign-Off

Name: DR. V. GURUPARAN.

Title: OneCare Clinic Management System.

Organization: OneCare Family Clinic

Signature: _____

Dr.V.Guruparan
MD (UKR) FID (UK)
SLMC No. 38242
Onecare Family Clinic

14.2 Project Supervisor Sign-Off

Name: _____

Date: _____

Signature: _____